Qualcomm

Open Source Summit Japan 2025

Status of DTS Validation in Linux Kernel

Krzysztof Kozlowski

Senior Staff Engineer
Qualcomm Wireless GmbH

krzk@kernel.org, @krzk@social.kernel.org

Introduction

- Krzysztof Kozlowski
- I work for Qualcomm

Introduction

- Krzysztof Kozlowski
- I work for Qualcomm
- I am a co-maintainer (with Rob Herring and Conor Dooley) of Devicetree bindings in the Linux kernel
  - I co-maintain also the soc subsystem (arm-soc)
  - … and a few Linux kernel subsystems and drivers

Introduction

- Krzysztof Kozlowski
- I work for Qualcomm
- I am a co-maintainer (with Rob Herring and Conor Dooley) of Devicetree bindings in the Linux kernel
    - I co-maintain also the soc subsystem (arm-soc)
    - … and a few Linux kernel subsystems and drivers
- I contributed a lot to the Linux kernel, although now I mostly read the code
    - git shortlog -s -n --no-merges | head -n 5

Agenda

- What is the DTS validation?
- Why does it matter (how many issues can this find?)
- The results!
  - Architectures
  - SoC Platforms

Disclaimer

- Presentation is based on actual data produced by Linux tools
    - Data tells whether given architecture or platform is compliant or not

Disclaimer

- Presentation is based on actual data produced by Linux tools
  - Data tells whether given architecture or platform is compliant or not
- However in a few slides I use adjectives like “incomplete” or “active” describing my opinion about given architecture or platform
- These opinions are my own and do not reflect the views of my employer

What Is the DTS Validation?

Brief Devicetree Introduction

- Devicetree: data format to describe the non-discoverable hardware
  - Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on

Brief Devicetree Introduction

- Devicetree: data format to describe the non-discoverable hardware
  - Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on
- DTS - Devicetree Sources are textual representation of Devicetree, to be consumed by the Devicetree compiler (DTC)

Brief Devicetree Introduction

- Devicetree: data format to describe the non-discoverable hardware
  - Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on
- DTS - Devicetree Sources are textual representation of Devicetree, to be consumed by the Devicetree compiler (DTC)
- DTB - Devicetree Blob is a compact binary representation of the DTS

Brief Devicetree Introduction

- Devicetree: data format to describe the non-discoverable hardware
  - Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on
- DTS - Devicetree Sources are textual representation of Devicetree, to be consumed by the Devicetree compiler (DTC)
- DTB - Devicetree Blob is a compact binary representation of the DTS
- Devicetree bindings - set of rules how DTS should be constructed and documenting the ABI between software and DTS

Brief Devicetree Introduction

- Devicetree: data format to describe the non-discoverable hardware
  - Used by the software, like Linux kernel or U-boot, to understand the hardware it is running on
- DTS - Devicetree Sources are textual representation of Devicetree, to be consumed by the Devicetree compiler (DTC)
- DTB - Devicetree Blob is a compact binary representation of the DTS
- Devicetree bindings - set of rules how DTS should be constructed and documenting the ABI between software and DTS
- Devicetree schema (DT Schema) is the current format of Devicetree bindings
  - … and the toolset
  - … and a project: https://github.com/devicetree-org/dt-schema/

What Is the DTS Validation?

- Devicetree bindings, in the DT Schema format, can be used to check if the DTS (sources) comply:
    - All required properties present
    - No undocumented ABI
    - All compatibles documented in the DT Schema format (not TXT!)
    - Property values within constraints

What Is the DTS Validation?

- Devicetree bindings, in the DT Schema format, can be used to check if the DTS (sources) comply:
  - All required properties present
  - No undocumented ABI
  - All compatibles documented in the DT Schema format (not TXT!)
  - Property values within constraints
- Compliance here means: no warnings from `make dtbs_check`

DTB (Target)

- Usually DTB represents one machine or board (as in hardware board)
  - Sometimes a flavor of it, e.g. one DTB per each development kit variant (MMC, UFS)

DTB (Target)

- Usually DTB represents one machine or board (as in hardware board)
    - Sometimes a flavor of it, e.g. one DTB per each development kit variant (MMC, UFS)
- DTB is also the Makefile target of each DTS
    - DTS => DTB

DTB (Target)

- Usually DTB represents one machine or board (as in hardware board)
    - Sometimes a flavor of it, e.g. one DTB per each development kit variant (MMC, UFS)
- DTB is also the Makefile target of each DTS
    - DTS => DTB
- Almost all overlays are applied in the Makefile to DTS as well
    - DTS + overlay => new DTB file

```
arch/arm64/boot/dts/qcom/Makefile:
dtb-$(CONFIG_ARCH_QCOM) += x1e80100-dell-xps13-9345.dtb x1e80100-dell-xps13-9345-el2.dtb
```

DTB (Target)

- Usually DTB represents one machine or board (as in hardware board)
  - Sometimes a flavor of it, e.g. one DTB per each development kit variant (MMC, UFS)
- DTB is also the Makefile target of each DTS
  - DTS => DTB
- Almost all overlays are applied in the Makefile to DTS as well
  - DTS + overlay => new DTB file

```
arch/arm64/boot/dts/qcom/Makefile:
dtb-$(CONFIG_ARCH_QCOM) += x1e80100-dell-xps13-9345.dtb x1e80100-dell-xps13-9345-el2.dtb
```

- Whenever mentioning here a “target”, it means DTB file, being created by the Makefile
  - Basically a `grep -E '(DTC|OVL).*\.dtb$'` on the output of `make dtbs_check`

Why Does It Matter?

How Many Issues Can This Find?

- In April 2024 I tried to evaluate the impact of DTS validation

How Many Issues Can This Find?

- In April 2024 I tried to evaluate the impact of DTS validation
- Presented results on Embedded Open Source Summit 2024:
  Whack-a-Mole with DTS Validation in the Linux Kernel
  Slides: https://sched.co/1aBEf
  Video recording: https://youtu.be/idkPl-d2lL8
- Rather rudimentary analysis yet still hopefully useful

How Many Issues Can This Find?

- In April 2024 I tried to evaluate the impact of DTS validation
- Presented results on Embedded Open Source Summit 2024:
  Whack-a-Mole with DTS Validation in the Linux Kernel
  Slides: https://sched.co/1aBEf
  Video recording: https://youtu.be/idkPl-d2lL8
- Rather rudimentary analysis yet still hopefully useful
- Evaluated Linux kernel commits for DTS if two SoC vendors:
  - Samsung Exynos, between Linux kernel v5.3 and v6.8
  - Qualcomm, between Linux kernel v5.18 and v6.8

Glossary (Still 2024 Recap)

- “Commit for DTS” means:
  - Touching arch/FOO/boot/dts/BAR/ files

Glossary (Still 2024 Recap)

- “Commit for DTS” means:
  - Touching arch/FOO/boot/dts/BAR/ files
- Commit being a result of DTS validation means:
  - Commit to fix explicit warning from DTS validation (`make dtbs_check`)
  - Usually clearly noted in the commit message
    - At least when people wrote good commit messages, explaining why they are doing something
      - … not what they are doing :(

Recap 2024 - Commits from the DTS Validation

- Of all commits touching DTS files in Linux kernel:
    - Samsung Exynos: 30-40% of commits were result of DTS validation
    - Qualcomm: 20-30% of commits were result of DTS validation

Results (Recap 2024)

This and several other slides are directly imported from my 2024 talk: https://sched.co/1aBEf

Platform	Total DTS commits	Commits as a result of DT schema validation	%
ARM Exynos v5.3 .. v6.8	370	152	41%
ARM64 Exynos v5.3 .. v6.8	139	46	33%
ARM Qualcomm v5.18 .. v6.8	476	142	30%
ARM64 Qualcomm v5.18 .. v6.8	2512	545	22%

Recap 2024 - Classification of Commits from DTS Validation

- Questions:
  - What types of issues are being flagged by DTS validation?
  - Are there real bugs being pointed out by DTS validation?

Recap 2024 - Classification of Commits from DTS Validation

- Questions:
  - What types of issues are being flagged by DTS validation?
  - Are there real bugs being pointed out by DTS validation?
- This was pretty subjective classification
  - Going through Samsung Exynos was a bit easier, because I mentioned warnings in the my commit messages
    - And there was not much other work in the subsystem happening

Recap 2024 - Classification of Commits from DTS Validation

- Questions:
  - What types of issues are being flagged by DTS validation?
  - Are there real bugs being pointed out by DTS validation?
- This was pretty subjective classification
  - Going through Samsung Exynos was a bit easier, because I mentioned warnings in the my commit messages
    - And there was not much other work in the subsystem happening
  - Many Qualcomm commits said nothing about warning, nothing about DT schema, but just word “bindings”
    - If only people could answer “why” they are doing what they are doing…

Recap 2024 - Classification of Commits from DTS Validation

- Questions:
  - What types of issues are being flagged by DTS validation?
  - Are there real bugs being pointed out by DTS validation?
- This was pretty subjective classification
  - Going through Samsung Exynos was a bit easier, because I mentioned warnings in the my commit messages
    - And there was not much other work in the subsystem happening
  - Many Qualcomm commits said nothing about warning, nothing about DT schema, but just word “bindings”
    - If only people could answer “why” they are doing what they are doing…
- See my slides from 2024 for the reasoning behind classification and examples

Classification: Groups (Recap 2024)

- Just to make DT schema happy
  - STYLE

Classification: Groups (Recap 2024)

- Just to make DT schema happy
  - STYLE
- Better hardware representation in DTS
  - IMPROVEMENT

Classification: Groups (Recap 2024)

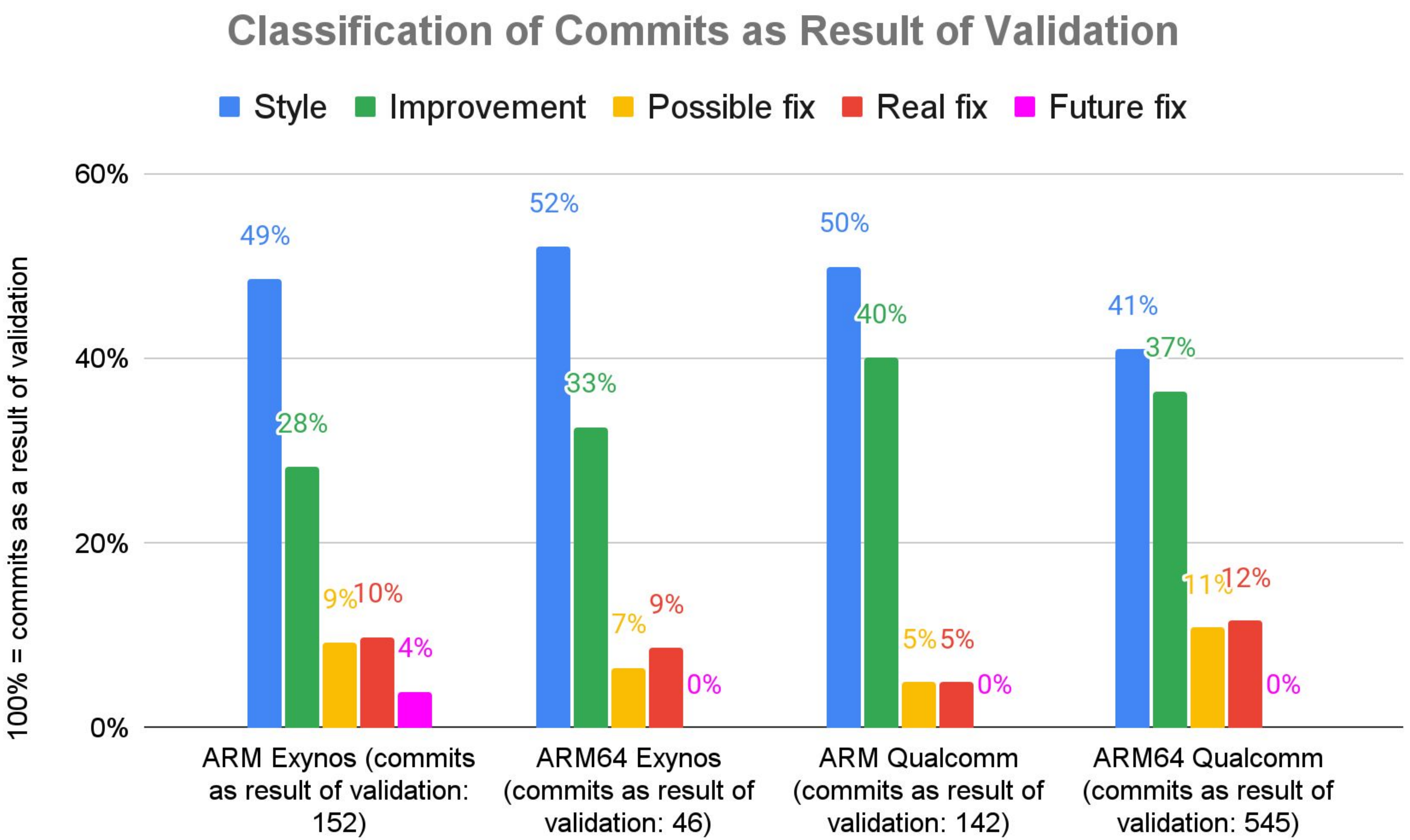
- Just to make DT schema happy
  - **STYLE**
- Better hardware representation in DTS
  - **IMPROVEMENT**
- Everything should work on Linux, but other DTS user (like U-Boot) could be affected
  - **POSSIBLE FIX**

Classification: Groups (Recap 2024)

- Just to make DT schema happy
  - STYLE
- Better hardware representation in DTS
  - IMPROVEMENT
- Everything should work on Linux, but other DTS user (like U-Boot) could be affected
  - POSSIBLE FIX
- Something does not work
  - REAL FIX

Classification: Groups (Recap 2024)

- Just to make DT schema happy
  - STYLE
- Better hardware representation in DTS
  - IMPROVEMENT
- Everything should work on Linux, but other DTS user (like U-Boot) could be affected
  - POSSIBLE FIX
- Something does not work
  - REAL FIX
- Something does not work, but DT schema did not detect it yet
  - FUTURE FIX


Classification of Commits as Result of Validation
Style
Improvement
Possible fix
Real fix
Future fix
60%
40%
20%
0%
100% = commits as a result of validation
49%
28%
9%
10%
4%
52%
33%
7%
9%
0%
50%
40%
5%
5%
0%
41%
37%
11%
12%
0%
ARM Exynos (commits as result of validation: 152)
ARM64 Exynos (commits as result of validation: 46)
ARM Qualcomm (commits as result of validation: 142)
ARM64 Qualcomm (commits as result of validation: 545)

Why Does It Matter?

- Because it points out real bugs in DTS (~10% of the cases)

Why Does It Matter?

- Because it points out real bugs in DTS (~10% of the cases)
- And makes DTS better in the rest of the cases

Why Does It Matter?

- Because it points out real bugs in DTS (~10% of the cases)
- And makes DTS better in the rest of the cases
- If better code did not convince you then:
  It is a requirement for Arm SystemReady Devicetree band

Status of Architectures

Kernel Versions

- All further slides show results as of Linux kernel:
  - Future v6.19 (next-20251125) - not for all slides
    - Validated with dtschema v2025.08
  - Latest v6.18, released few days ago
    - Validated with dtschema v2025.08
  - v6.13, released January 2025
    - Validated with dtschema v2024.11
  - v6.7, released January 2024
    - Validated with dtschema v2024.02

Kernel Versions

- All further slides show results as of Linux kernel:
  - Future v6.19 (next-20251125) - not for all slides
    - Validated with dtschema v2025.08
  - Latest v6.18, released few days ago
    - Validated with dtschema v2025.08
  - v6.13, released January 2025
    - Validated with dtschema v2024.11
  - v6.7, released January 2024
    - Validated with dtschema v2024.02
- Trying to show change happening during last year

Kernel Versions

- All further slides show results as of Linux kernel:
  - Future v6.19 (next-20251125) - not for all slides
    - Validated with dtschema v2025.08
  - Latest v6.18, released few days ago
    - Validated with dtschema v2025.08
  - v6.13, released January 2025
    - Validated with dtschema v2024.11
  - v6.7, released January 2024
    - Validated with dtschema v2024.02
- Trying to show change happening during last year
- Comparing with earlier Linux kernel releases is tricky, because dtschema is stricter now
  - Therefore trying to use dtschema package from that era

Architectures Using DTS

- There are 13 architectures in the Linux kernel using DTS files
  - `ls arch/*/boot/dts`

Architectures Using DTS

- There are 13 architectures in the Linux kernel using DTS files
  - `ls arch/*/boot/dts`
- Most of them legacy and not frequently updated
  - PowerPC:
    - Still updated, but little work is done to make it compliant with checks
    - A lot of legacy stuff, predating current DT (ePAPR)

Architectures Using DTS

- There are 13 architectures in the Linux kernel using DTS files
  - `ls arch/*/boot/dts`
- Most of them legacy and not frequently updated
  - PowerPC:
    - Still updated, but little work is done to make it compliant with checks
    - A lot of legacy stuff, predating current DT (ePAPR)
  - MIPS
    - Alive, but somehow not easy to work with - impossible to build all DTBs at once (allyesconfig, COMPILE_TEST), issue being fixed by Rob

Architectures Using DTS

- There are 13 architectures in the Linux kernel using DTS files
  - `ls arch/*/boot/dts`
- Most of them legacy and not frequently updated
  - PowerPC:
    - Still updated, but little work is done to make it compliant with checks
    - A lot of legacy stuff, predating current DT (ePAPR)
  - MIPS
    - Alive, but somehow not easy to work with - impossible to build all DTBs at once (allyesconfig, COMPILE_TEST), issue being fixed by Rob
  - Several other (ARC, Microblaze, NIOS2) are basically legacy and no work is done
  - No point to measure legacy

Architectures Using DTS

- There are 13 architectures in the Linux kernel using DTS files
  - `ls arch/*/boot/dts`
- Most of them legacy and not frequently updated
  - PowerPC:
    - Still updated, but little work is done to make it compliant with checks
    - A lot of legacy stuff, predating current DT (ePAPR)
  - MIPS
    - Alive, but somehow not easy to work with - impossible to build all DTBs at once (allyesconfig, COMPILE_TEST), issue being fixed by Rob
  - Several other (ARC, Microblaze, NIOS2) are basically legacy and no work is done
  - No point to measure legacy
- Some architecture or SoC platforms have both ACPI and Devicetree, but only ACPI is being actively developed
  - Lack of activity in DTS does not necessarily mean stuff is definitely abandoned!

Architectures Tested Here

- ARM
  - `ls arch/arm/boot/dts`
  - Everything which could build, so includes ARMv7, ARMv6 and ancient ARMv4T (ARM9 like TI Nspire calculator)
- ARM64
  - `ls arch/arm64/boot/dts`

Architectures Tested Here

- ARM
  - `ls arch/arm/boot/dts`
  - Everything which could build, so includes ARMv7, ARMv6 and ancient ARMv4T (ARM9 like TI Nspire calculator)
- ARM64
  - `ls arch/arm64/boot/dts`
- RISC-V
- Loongarch
- PowerPC

Architectures Tested Here

- ARM
  - `ls arch/arm/boot/dts`
  - Everything which could build, so includes ARMv7, ARMv6 and ancient ARMv4T (ARM9 like TI Nspire calculator)
- ARM64
  - `ls arch/arm64/boot/dts`
- RISC-V
- Loongarch
- PowerPC
- Not included in my talk
  - MIPS - allyesconfig builds only part of it (being fixed by Rob)
  - Pity if some architectures wanted to be on this list, but `git log` does not lie

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
- Is there an architecture with all targets failing (nothing warning-free)?

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
- Is there an architecture with all targets failing (nothing warning-free)?
  - Yes!

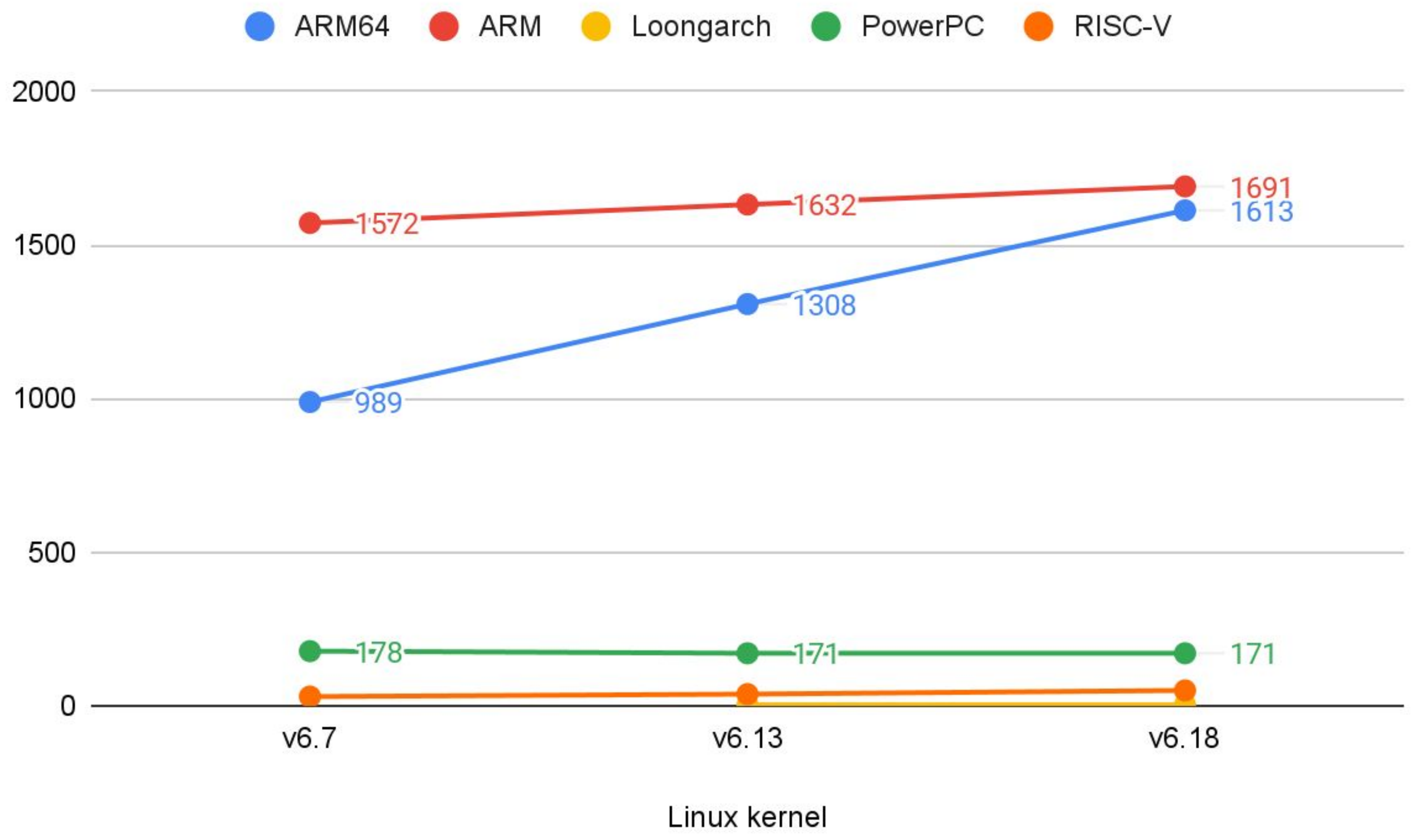
Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
- Is there an architecture with all targets failing (nothing warning-free)?
  - Yes!
- Same issue in a DTSI file, included by multiple DTS, will result in multiple warnings

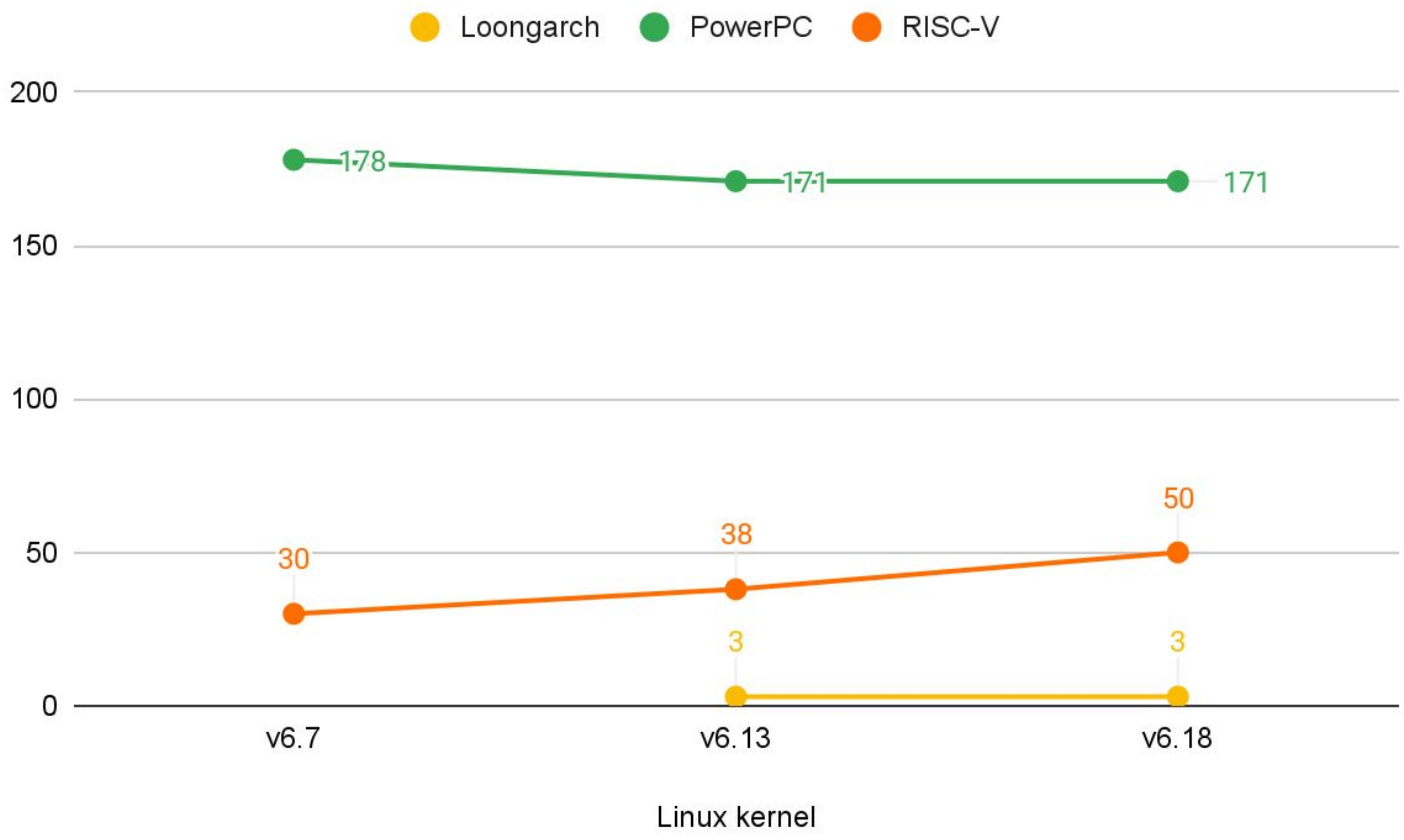
Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
    - Yes!
- Is there an architecture with all targets failing (nothing warning-free)?
    - Yes!
- Same issue in a DTSI file, included by multiple DTS, will result in multiple warnings
    - To offset this, more interesting metrics are:
        - number of warnings / number of targets
        - number of warnings / number of Lines of Code (kLoC)

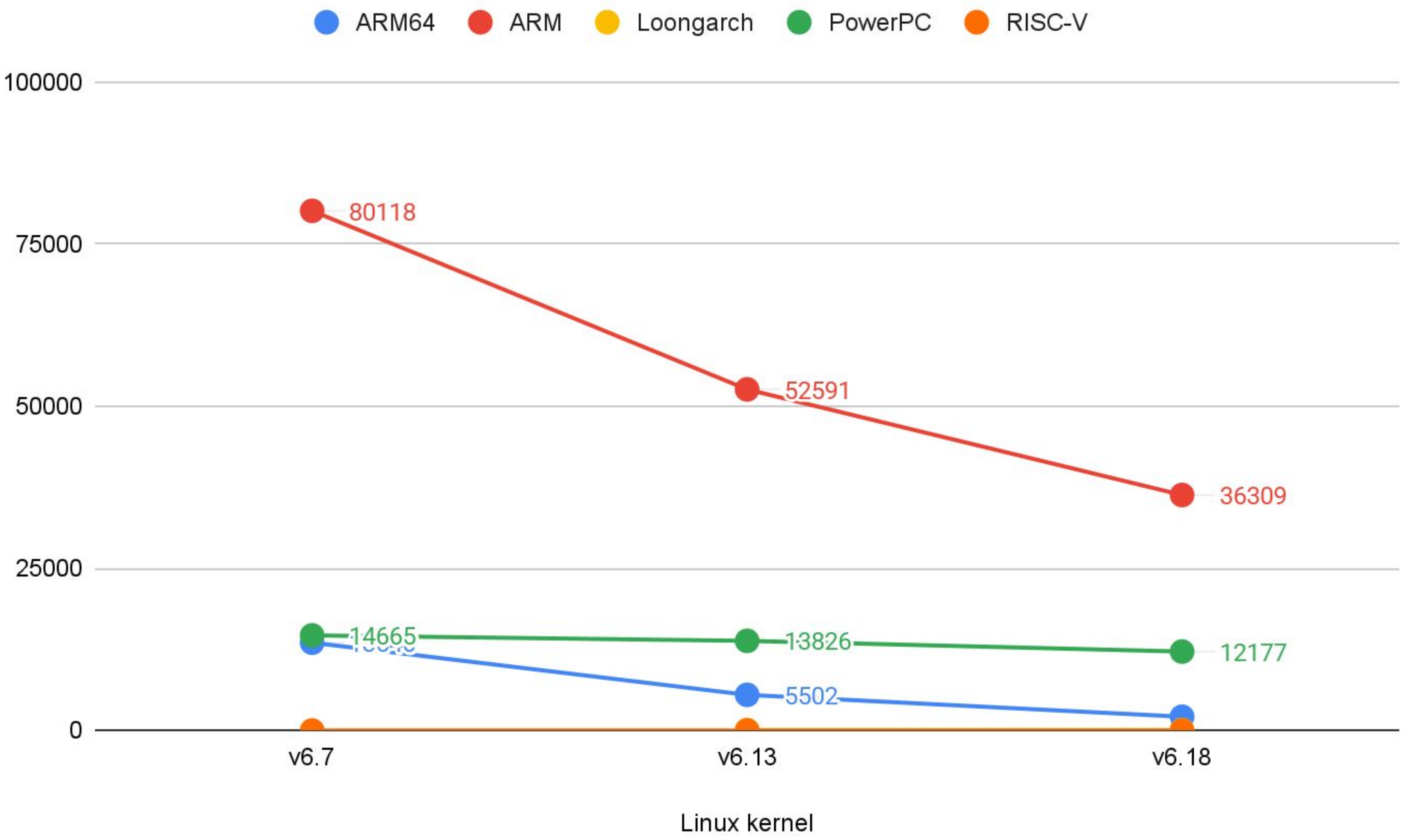
Targets per Architecture



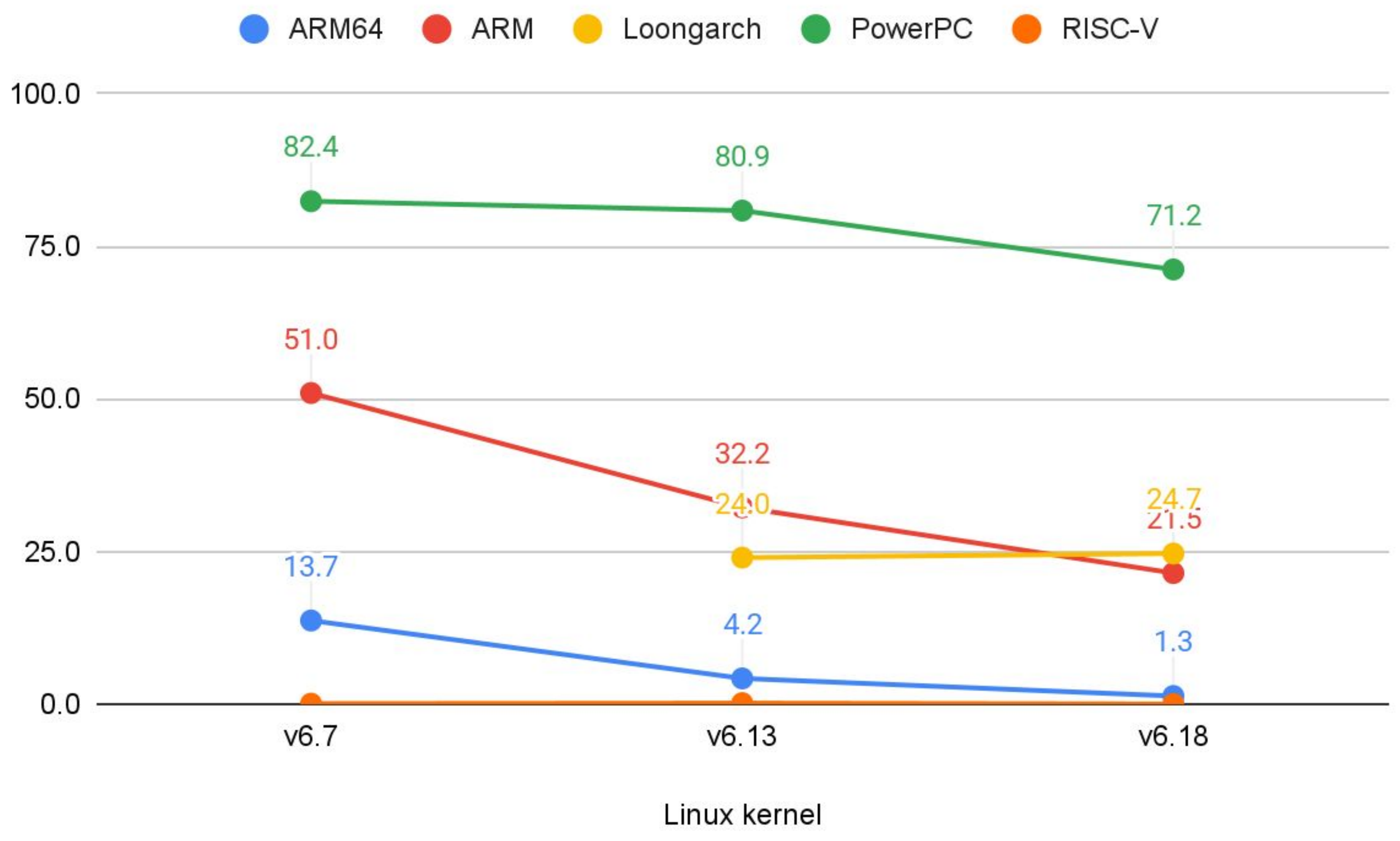
Targets per Architecture (Loongarch, PowerPC and RISC-V)



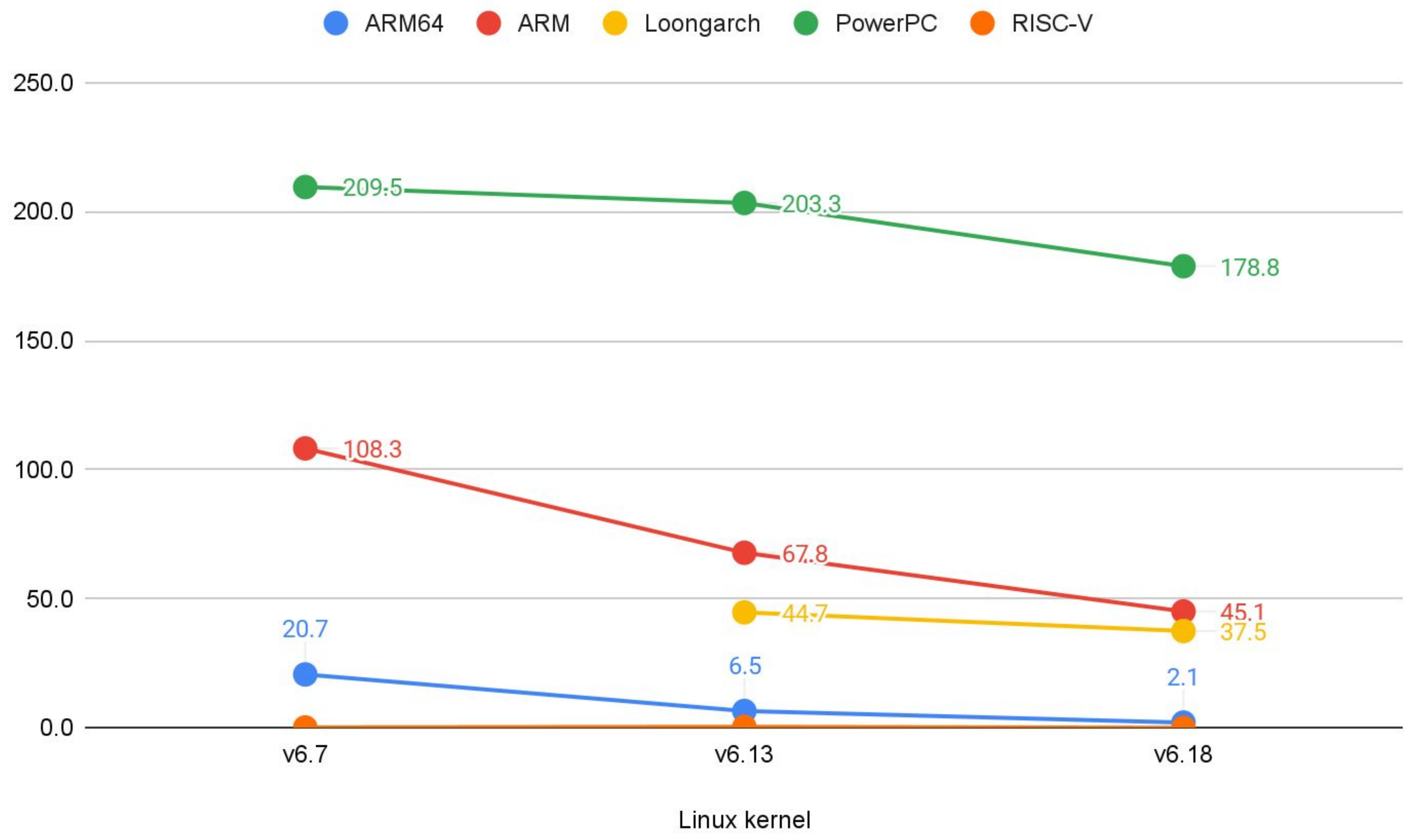
Total DTBs Check Warnings (Lower Better)



DTBs Check Warnings / Target (Lower Better)



DTBs Check Warnings / 1000-Lines-of-Code (Lower Better)



DTBs Check Warnings / Target (Lower Better)

Linux kernel	ARM64	ARM	Loongarch	PowerPC	RISC-V
v6.7	13.7	51.0	N/A	82.4	0.1
v6.13	4.2	32.2	24.0	80.9	0.2
v6.18	1.3	21.5	24.7	71.2	0.0

DTBs Check Warnings / Target (Lower Better)

Linux kernel	ARM64	ARM	Loongarch	PowerPC	RISC-V
v6.7	13.7	51.0	N/A	82.4	0.1
v6.13	4.2	32.2	24.0	80.9	0.2
v6.18	1.3	21.5	24.7	71.2	0.0

- Loongarch has only three DTB targets and a lot of warnings per target

DTBs Check Warnings / Target (Lower Better)

Linux kernel	ARM64	ARM	Loongarch	PowerPC	RISC-V
v6.7	13.7	51.0	N/A	82.4	0.1
v6.13	4.2	32.2	24.0	80.9	0.2
v6.18	1.3	21.5	24.7	71.2	0.0

- Loongarch has only three DTB targets and a lot of warnings per target
- RISC-V has fifty DTB target and no warnings
  - RISC-V was actually warnings free for two or more years, with minor occasional bumps in the warnings

DTBs Check Warnings / Target (Lower Better)

Linux kernel	ARM64	ARM	Loongarch	PowerPC	RISC-V
v6.7	13.7	51.0	N/A	82.4	0.1
v6.13	4.2	32.2	24.0	80.9	0.2
v6.18	1.3	21.5	24.7	71.2	0.0

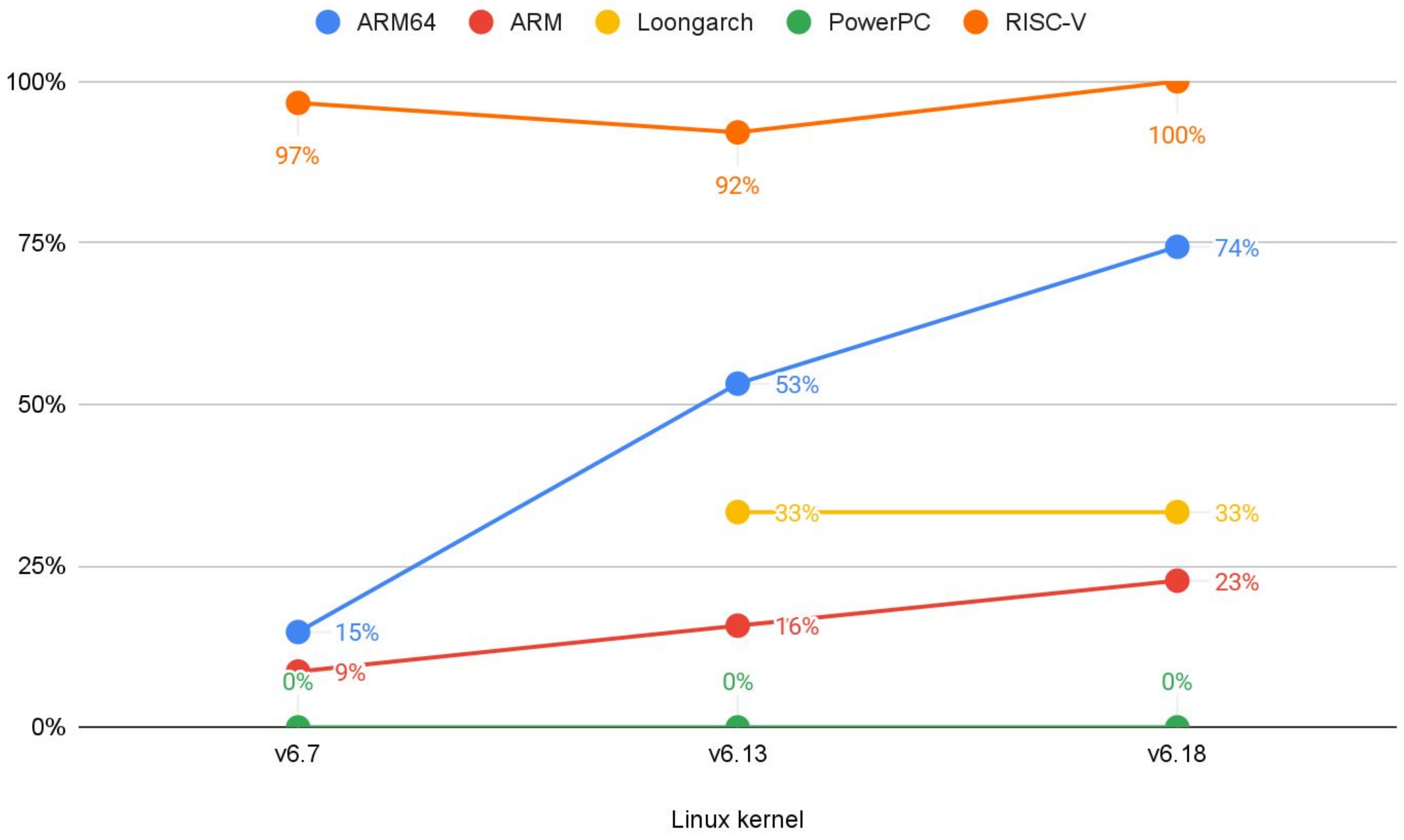
- Loongarch has only three DTB targets and a lot of warnings per target
- RISC-V has fifty DTB target and no warnings
  - RISC-V was actually warnings free for two or more years, with minor occasional bumps in the warnings
- ARM and ARM64 has comparable size (number of targets) but different number of warnings

DTBs Check Warnings / Target (Lower Better)

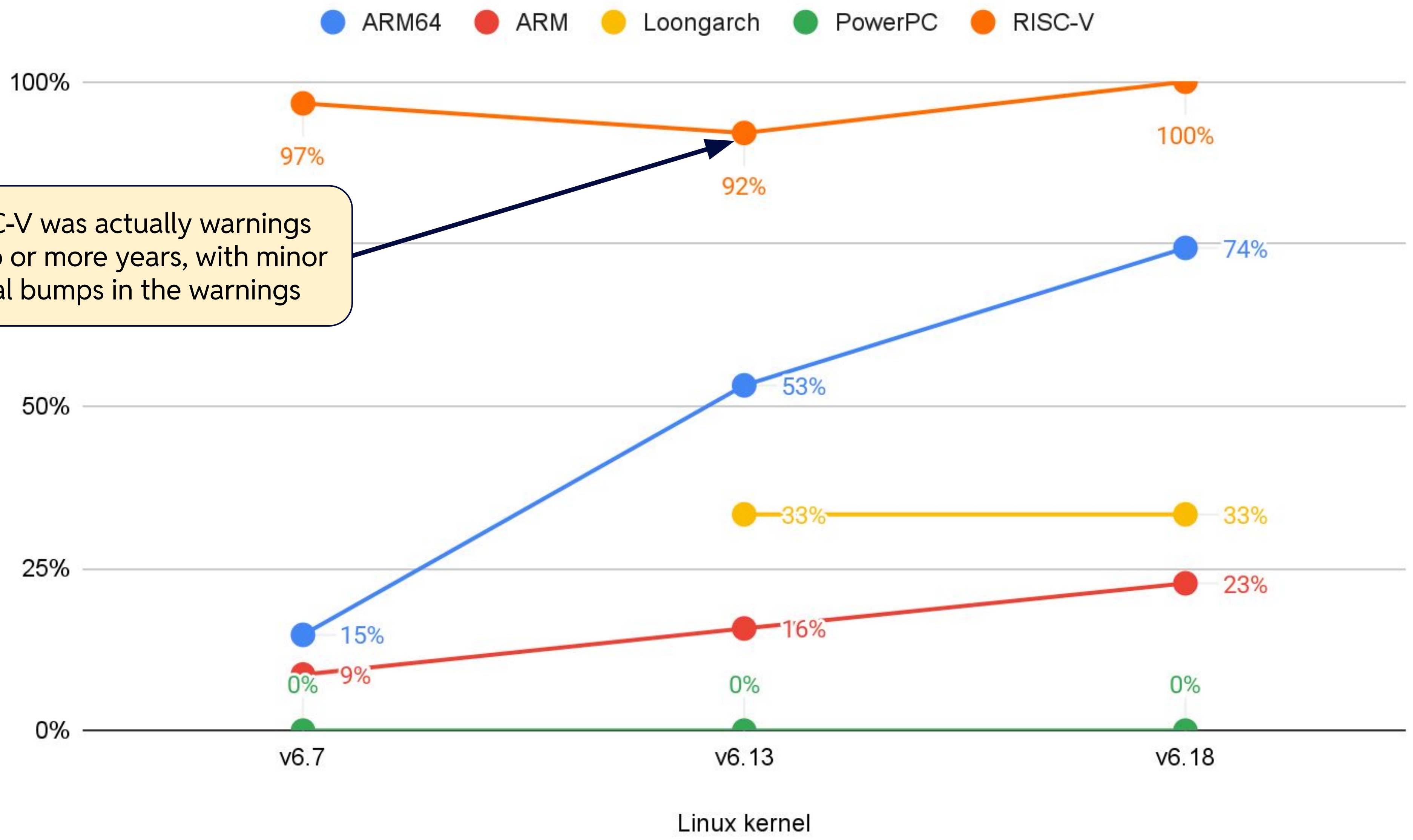
Linux kernel	ARM64	ARM	Loongarch	PowerPC	RISC-V
v6.7	13.7	51.0	N/A	82.4	0.1
v6.13	4.2	32.2	24.0	80.9	0.2
v6.18	1.3	21.5	24.7	71.2	0.0

- Loongarch has only three DTB targets and a lot of warnings per target
- RISC-V has fifty DTB target and no warnings
  - RISC-V was actually warnings free for two or more years, with minor occasional bumps in the warnings
- ARM and ARM64 has comparable size (number of targets) but different number of warnings
- Conclusion: the size of your ecosystem does not guarantee your DTS will be correct
  - Thank you, Captain Obvious!

Percentage of Warning-free Targets (Higher Better)

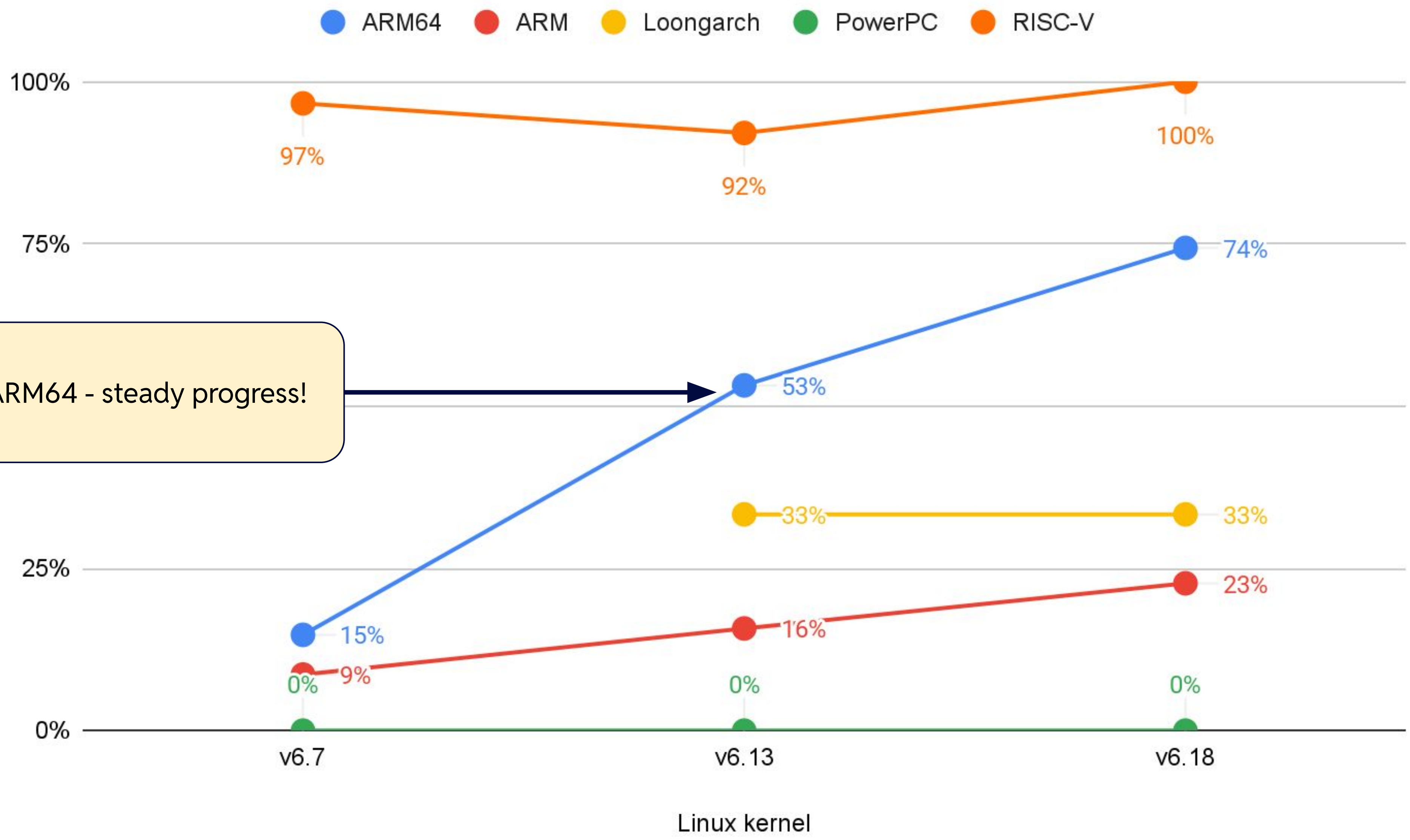


Percentage of Warning-free Targets (Higher Better)



Note: RISC-V was actually warnings free for two or more years, with minor occasional bumps in the warnings

Percentage of Warning-free Targets (Higher Better)



Good job ARM64 - steady progress!

Commentary

- New, fresh architectures without legacy have it easier

Commentary

- New, fresh architectures without legacy have it easier
  - If they care…
  - See Loongarch, a fresh architecture with big amount of warnings and no one caring
    - Only 1 out 3 DTBs clean of warnings…

Commentary

- New, fresh architectures without legacy have it easier
  - If they care…
  - See Loongarch, a fresh architecture with big amount of warnings and no one caring
    - Only 1 out 3 DTBs clean of warnings…
- Old architectures with “stable” boards just don’t get fixes?

Commentary

- New, fresh architectures without legacy have it easier
  - If they care…
  - See Loongarch, a fresh architecture with big amount of warnings and no one caring
    - Only 1 out 3 DTBs clean of warnings…
- Old architectures with “stable” boards just don’t get fixes?
- The biggest architecture in the kernel - ARM64 - can get fixed as well!

Status of (SoCs) Platforms

SoC Platforms

- Platforms are collection of SoCs or machines coming from one vendor, usually maintained by dedicated maintainer(s)

SoC Platforms

- Platforms are collection of SoCs or machines coming from one vendor, usually maintained by dedicated maintainer(s)
- Mostly organized in subdirectories, e.g.:
  - Qualcomm:
    - arch/arm/boot/dts/qcom/
    - arch/arm64/boot/dts/qcom/

SoC Platforms

- Platforms are collection of SoCs or machines coming from one vendor, usually maintained by dedicated maintainer(s)
- Mostly organized in subdirectories, e.g.:
  - Qualcomm:
    - arch/arm/boot/dts/qcom/
    - arch/arm64/boot/dts/qcom/
  - Microchip:
    - arch/arm/boot/dts/microchip/
    - arch/arm64/boot/dts/microchip/
    - arch/riscv/boot/dts/microchip/
  - But there are exceptions…

SoC Platforms

- Platforms are collection of SoCs or machines coming from one vendor, usually maintained by dedicated maintainer(s)
- Mostly organized in subdirectories, e.g.:
    - Qualcomm:
        - arch/arm/boot/dts/qcom/
        - arch/arm64/boot/dts/qcom/
    - Microchip:
        - arch/arm/boot/dts/microchip/
        - arch/arm64/boot/dts/microchip/
        - arch/riscv/boot/dts/microchip/
    - But there are exceptions…
- Sometimes very different SoC families within, e.g. Freescale LS and NXP i.MX
    - Five subdirectories in arch/arm/boot/dts/nxp/

SoC Platforms Being Actively Improved

- Sometimes big difference in the activity and size:
  - git log --no-merges --oneline v6.7..v6.18 -- arch/arm64/boot/dts/**qcom**/ | wc -l
    - 1982
  - git ls-files arch/arm64/boot/dts/qcom/ | wc -l
    - 516
  - git log --no-merges --oneline v6.7..v6.18 -- arch/arm64/boot/dts/**intel**/ | wc -l
    - 22
  - git ls-files arch/arm64/boot/dts/intel/ | wc -l
    - 11

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support
  - Small DTSI/DTS

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support
  - Small DTSI/DTS
  - Only some of devices described

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support
  - Small DTSI/DTS
  - Only some of devices described
  - It is unfair to compare an SoC with a DTSI of ~300 lines of code (synaptics/berlin4ct.dtsi) and one with 8300 lines (qcom/sm8650.dtsi)

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support
  - Small DTSI/DTS
  - Only some of devices described
  - It is unfair to compare an SoC with a DTSI of ~300 lines of code (synaptics/berlin4ct.dtsi) and one with 8300 lines (qcom/sm8650.dtsi)
- Reminder:
  Same issue in a DTSI file, included by multiple DTS, will result in multiple warnings

Completeness of the SoC Platforms

- There are several platforms, e.g. in ARM and ARM64, which have incomplete upstream support
  - Small DTSI/DTS
  - Only some of devices described
  - It is unfair to compare an SoC with a DTSI of ~300 lines of code (synaptics/berlin4ct.dtsi) and one with 8300 lines (qcom/sm8650.dtsi)
- Reminder:
  Same issue in a DTSI file, included by multiple DTS, will result in multiple warnings
  - To offset this, more interesting metrics are:
    - number of warnings / number of targets
    - number of warnings / number of Lines of Code (kLoC)

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
    - Yes! More than one...

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one...
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one...
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
- Is there an incomplete platform with a very limited/small DTS and still not compliant at all?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one...
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
- Is there an incomplete platform with a very limited/small DTS and still not compliant at all?
  - Yes!

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
- Is there an incomplete platform with a very limited/small DTS and still not compliant at all?
  - Yes!
- Is there a platform with all targets failing (nothing warning-free)?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes! More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
- Is there an incomplete platform with a very limited/small DTS and still not compliant at all?
  - Yes!
- Is there a platform with all targets failing (nothing warning-free)?
  - Yes! More than one…

SoC Platforms - PowerPC

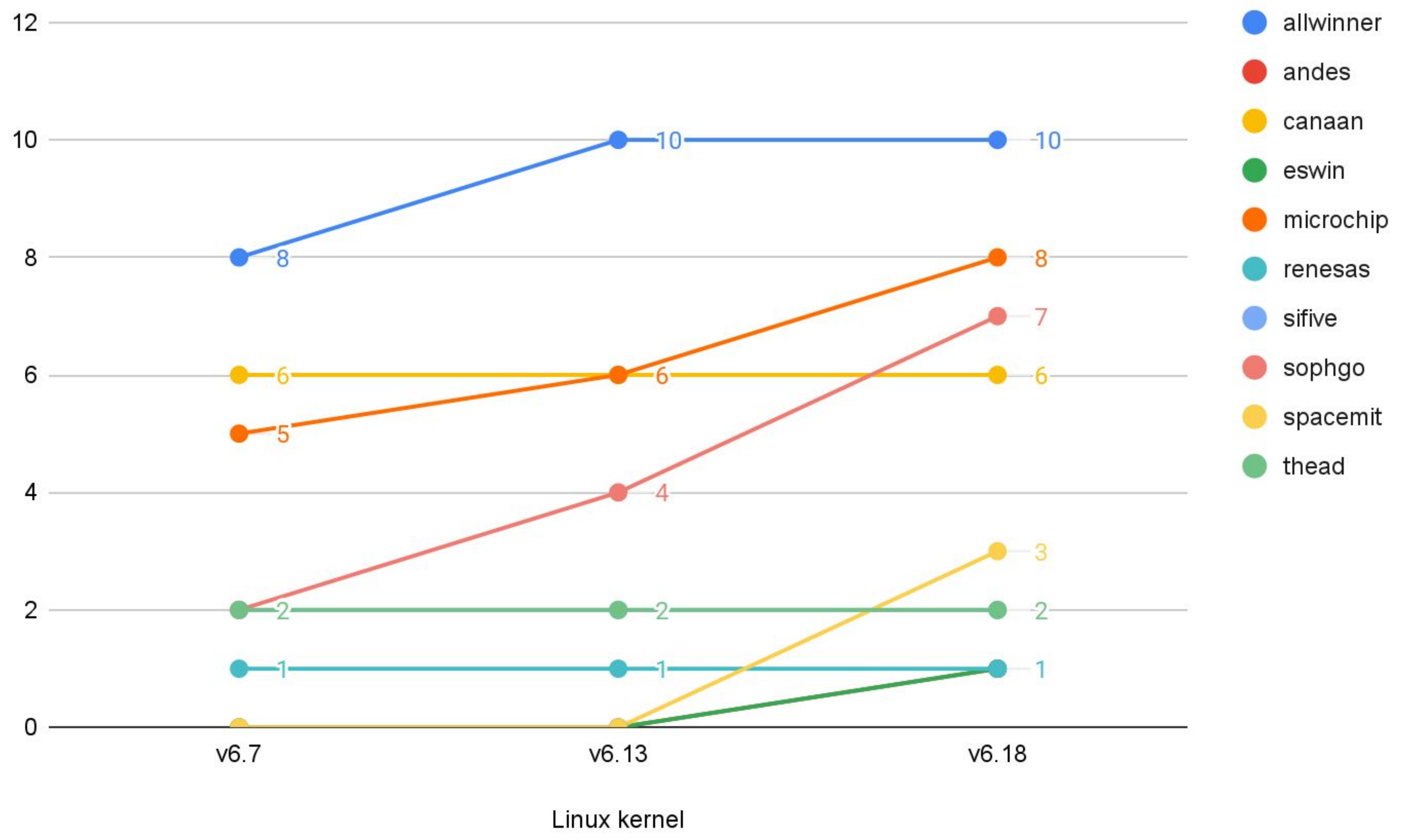
- PowerPC has no per-vendor split, thus not investigating individual platforms

Status of RISC-V Platforms

SoC Platforms - RISC-V

- RISC-V has 100% DTBs check compliance so no need to compare SoCs
- I can show number of targets…

Number of Targets - RISC-V

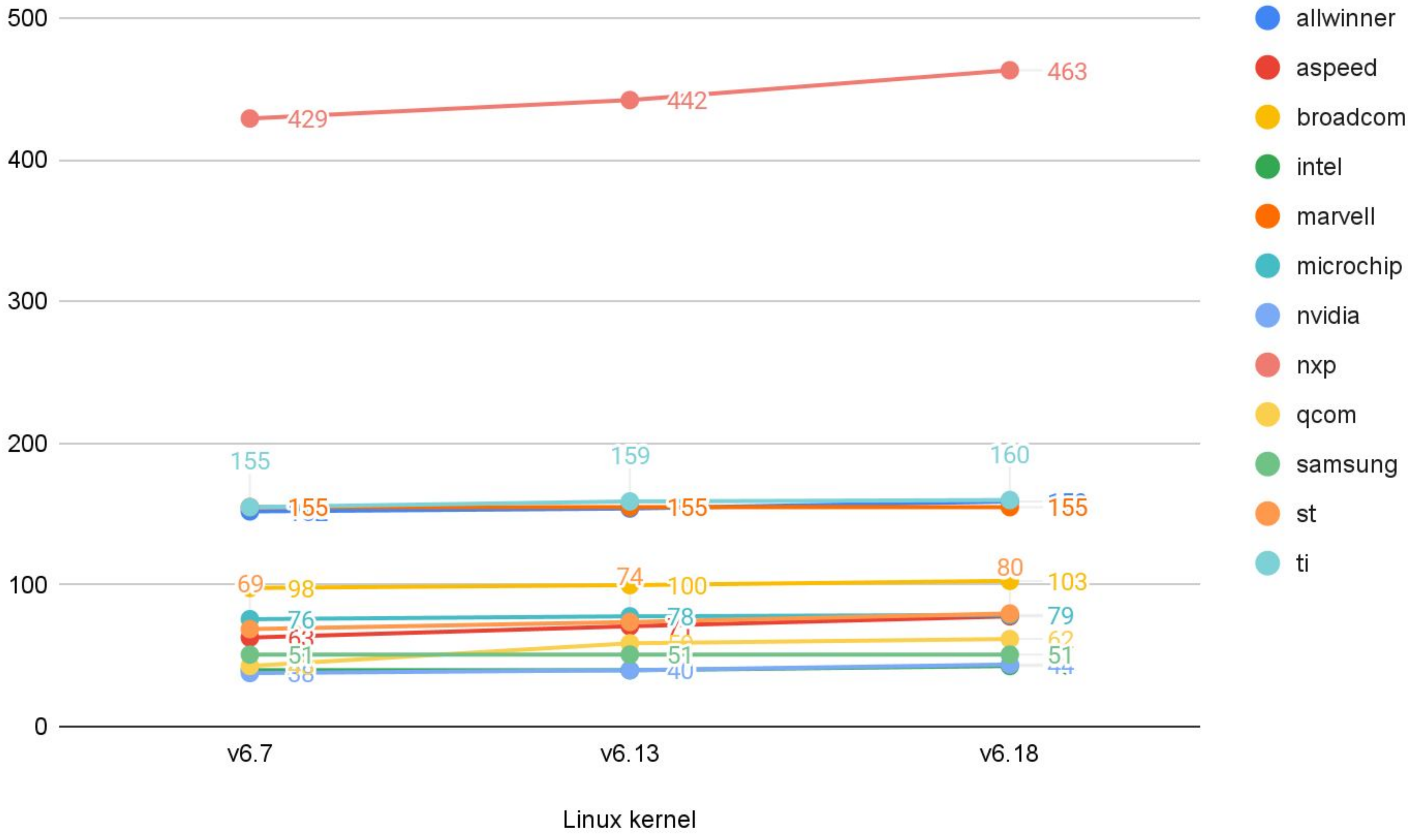


Status of ARM (32-bit) Platforms

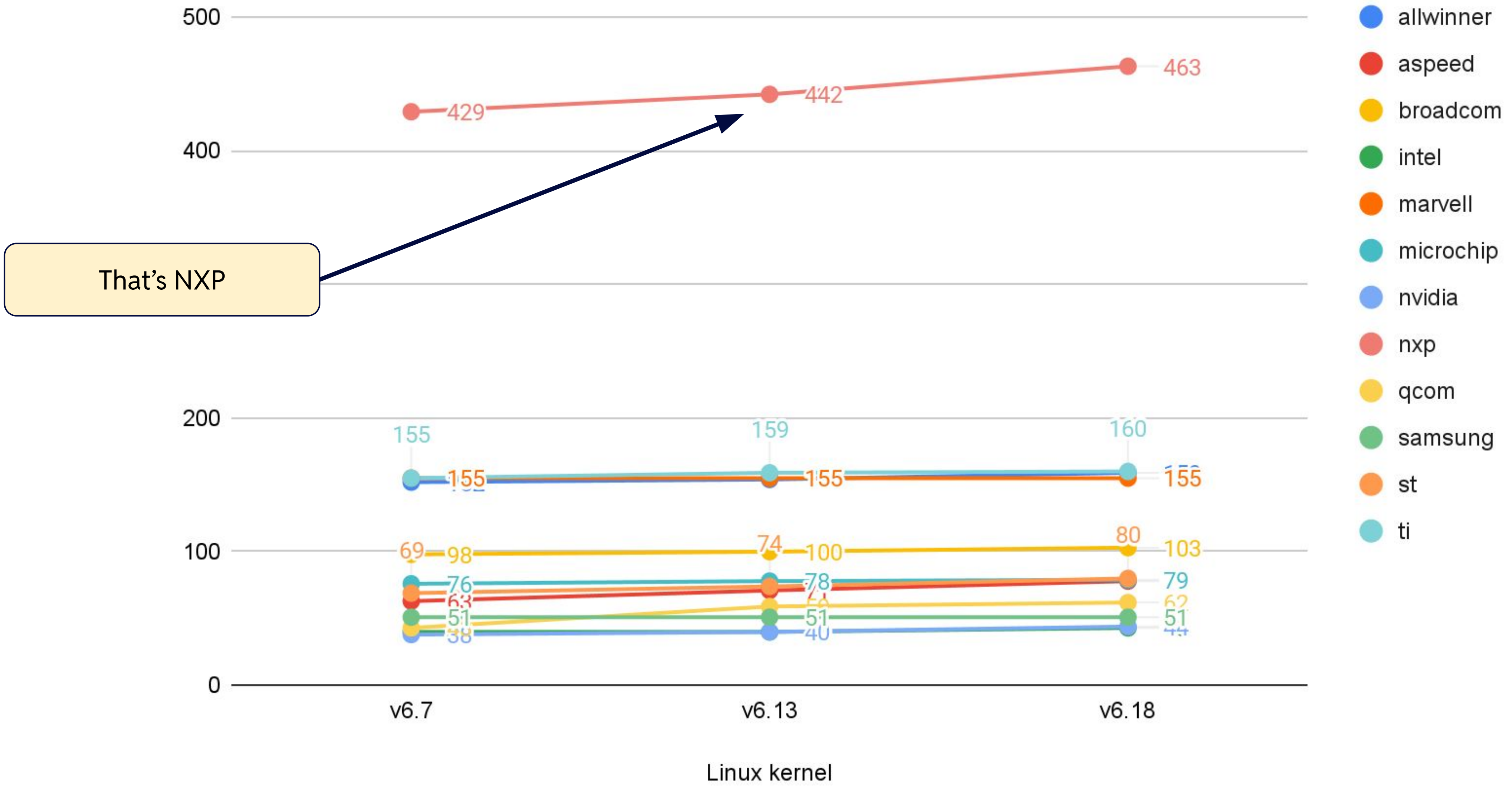
SoC Platforms - ARM

- There is 40 subdirectories in arch/arm/boot/dts - ~40 SoC vendors
- Many not being actively developed
- Following slides show subset of architectures, e.g. most active in Linux kernel

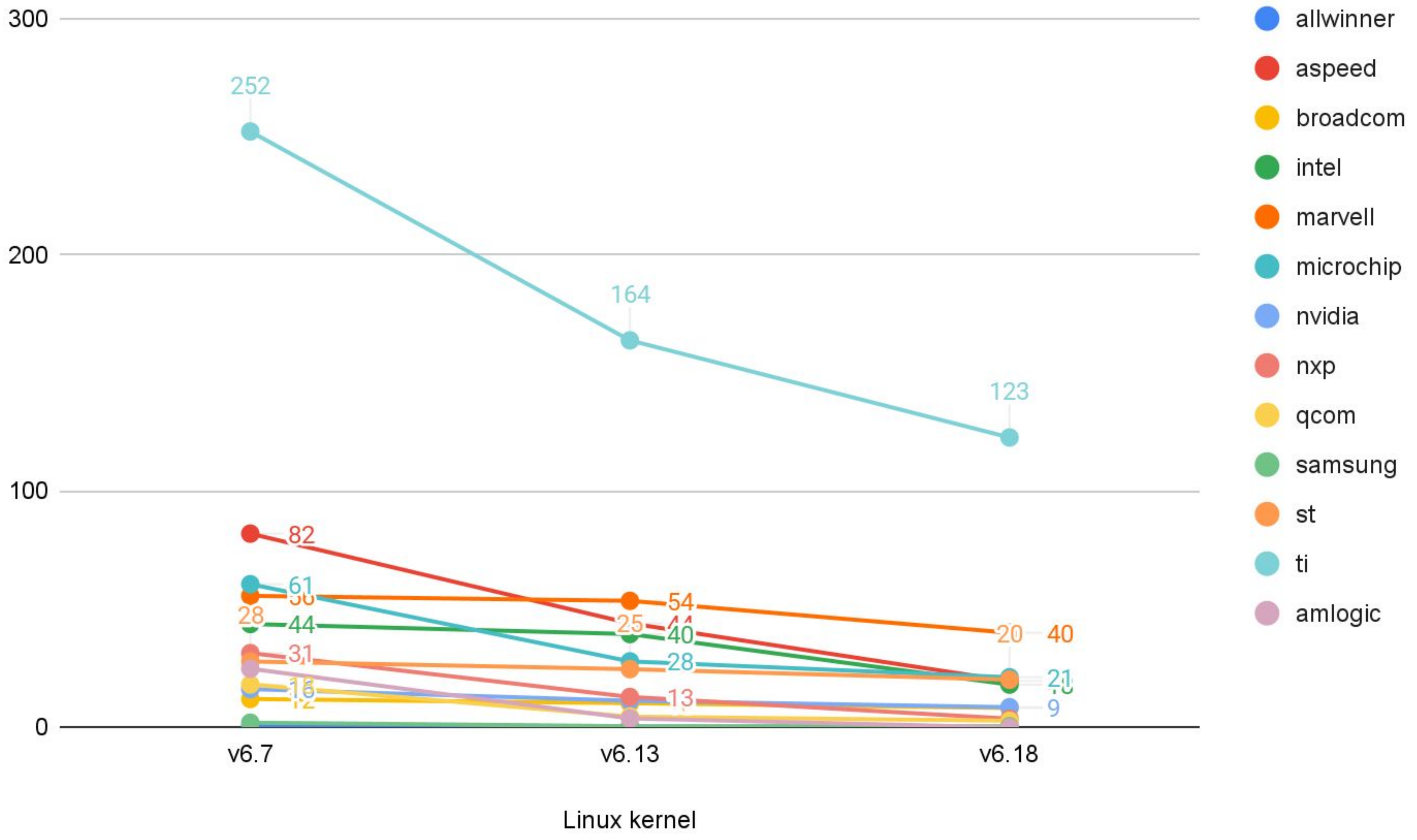
Number of Targets - ARM (Excerpt)



Number of Targets - ARM (Excerpt)

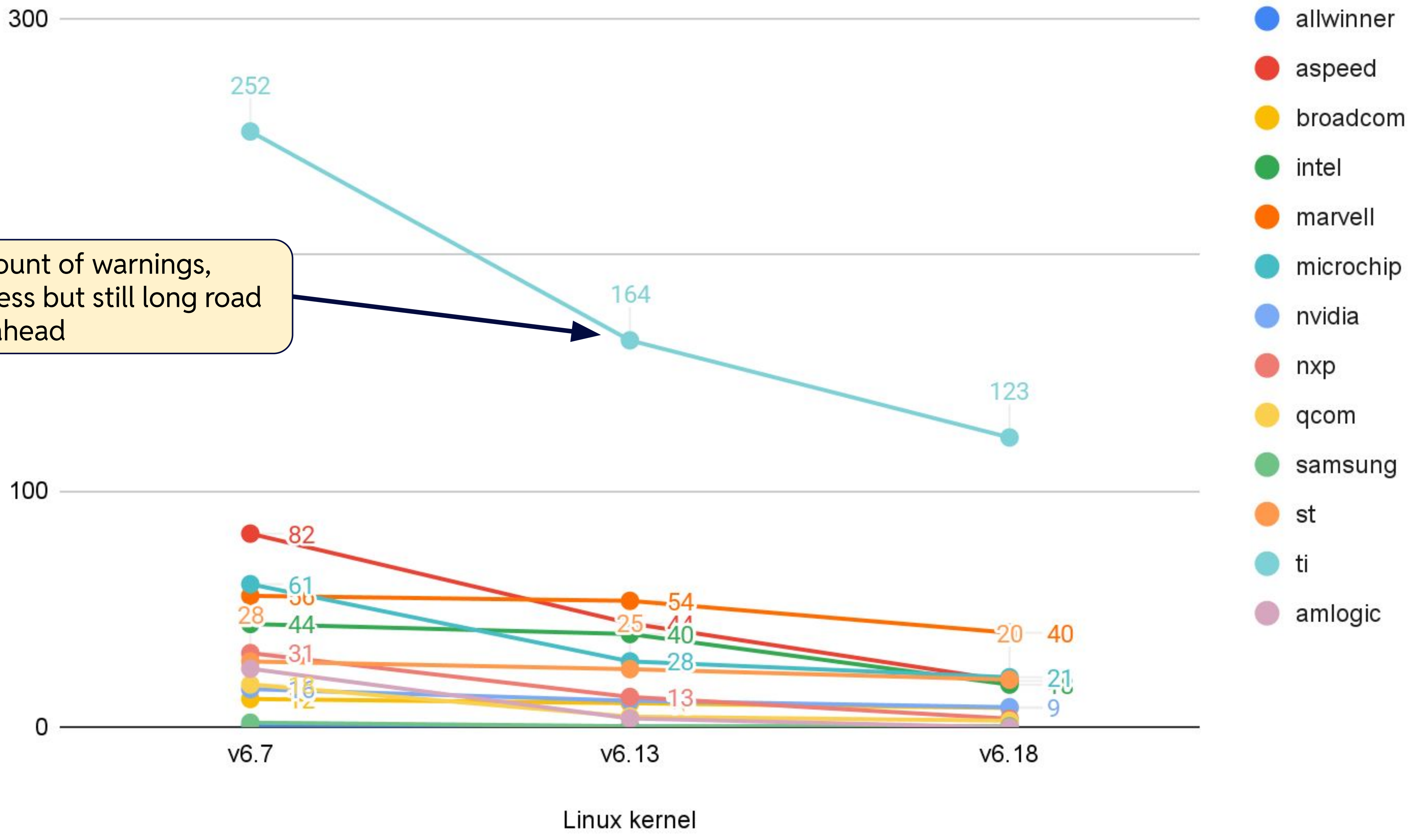


DTBs Check Warnings / Target (Lower Better) - ARM (Excerpt)

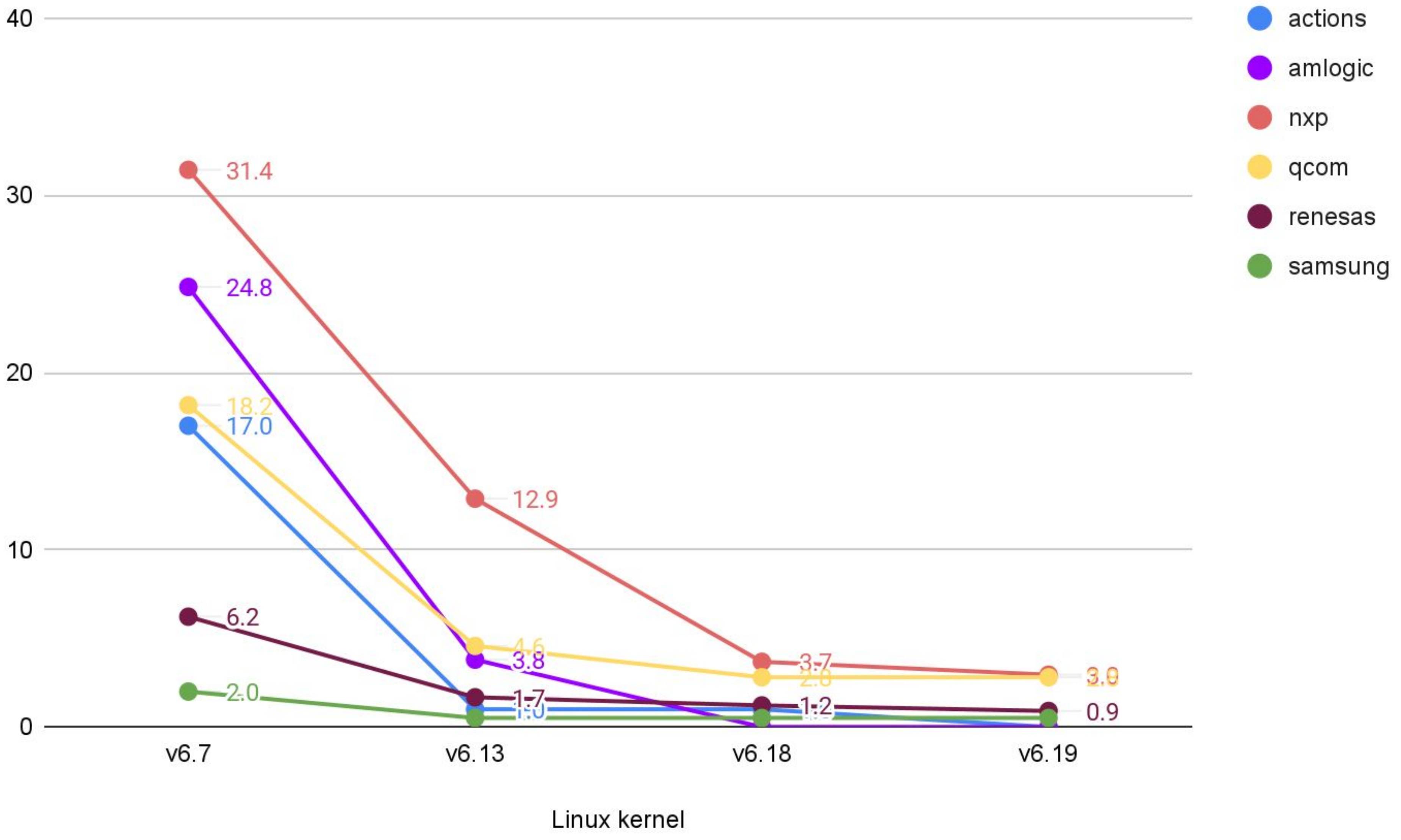


DTBs Check Warnings / Target (Lower Better) - ARM (Excerpt)

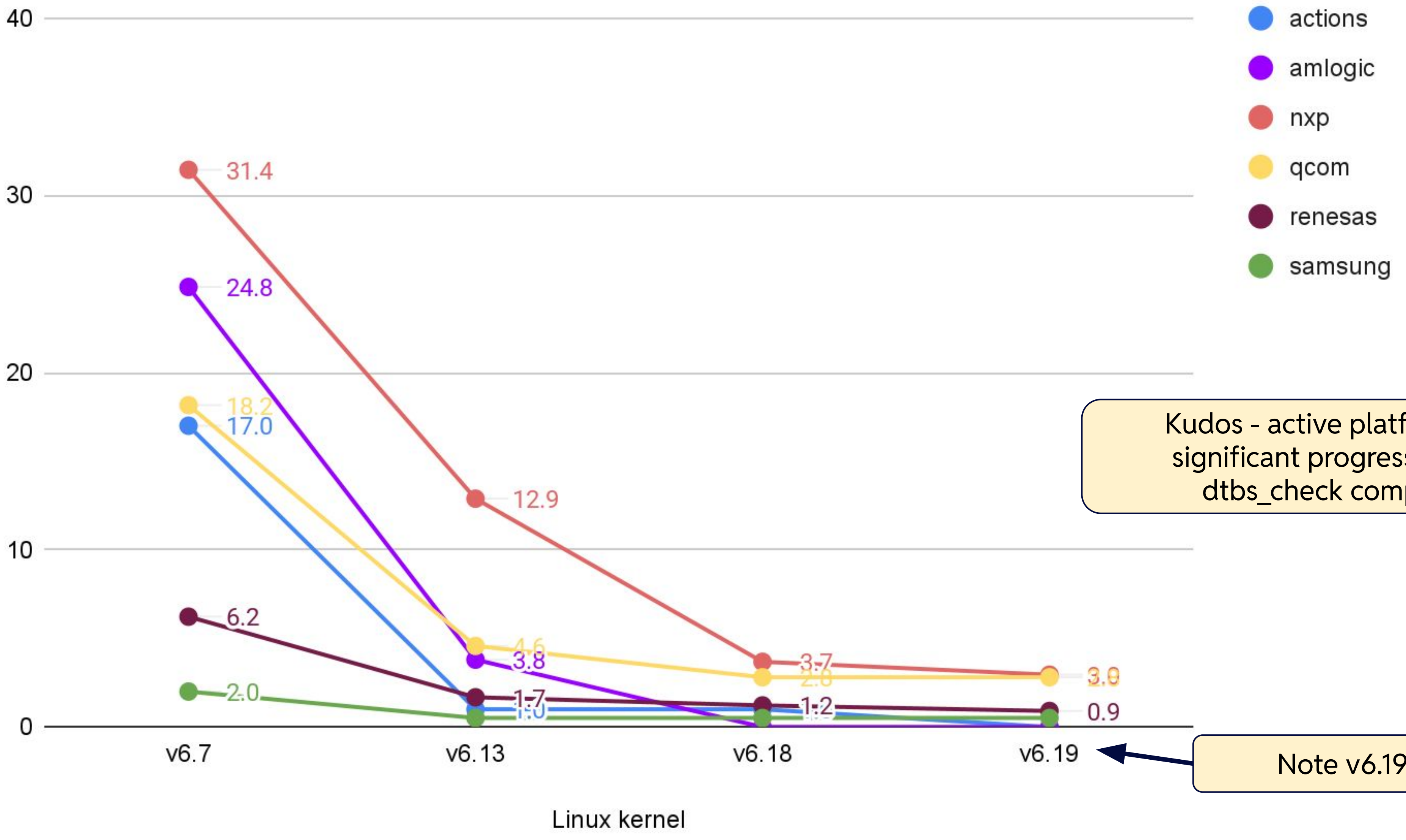
TI: Huge amount of warnings, significant progress but still long road ahead



DTBs Check Warnings / Target (Lower Better) - ARM Kudos



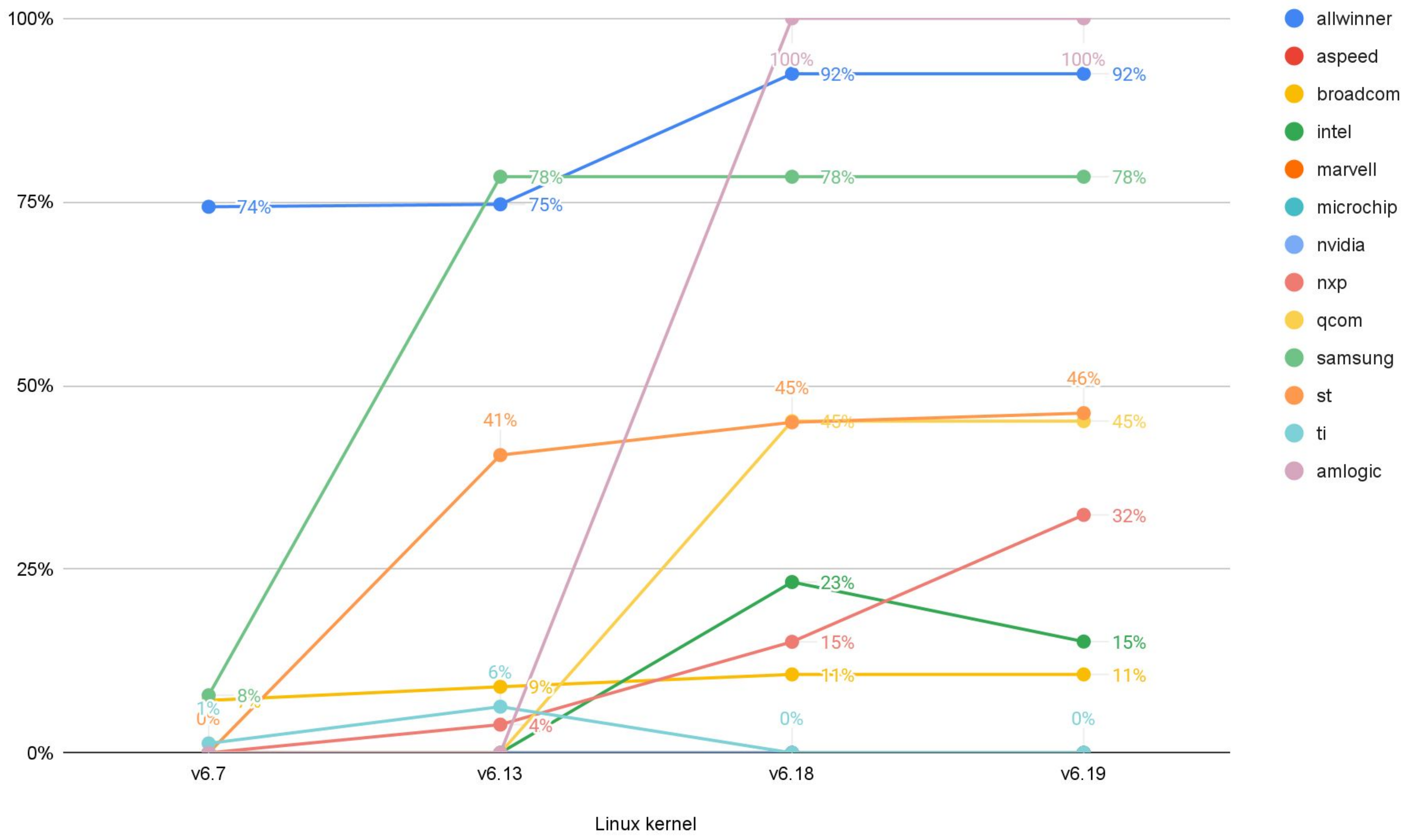
DTBs Check Warnings / Target (Lower Better) - ARM Kudos



Kudos - active platforms with significant progress towards dtbs_check compliance

Note v6.19 appearing

Percentage of Warning-free Targets (Higher Better) - ARM (Excerpt)



Warning-free Platforms (Higher Better) - ARM, v6.19 (next)

SoC Platform	Percentage of Warning-free Targets	No. Targets
Actions	100%	5
Airoha	100%	1
Amlogic	100%	6
HPE	100%	1
Allwinner	92.5%	159

Platforms with Only Warnings - ARM, v6.19 (next)

SoC Platform	Percentage of Warning-free Targets	No. Targets
alphascale	0%	1
amazon	0%	1
arm	0%	24
aspeed	0%	80
axis	0%	1
calxeda	0%	2
cnxt	0%	1
gemini	0%	10
marvell	0%	155
microchip	0%	79
moxa	0%	1
nspire	0%	3

SoC Platform	Percentage of Warning-free Targets	No. Targets
nuvoton	0%	6
nvidia	0%	45
realtek	0%	2
sigmastar	0%	8
sunplus	0%	1
synaptics	0%	4
ti	0%	161
unisoc	0%	2
vt8500	0%	6
xen	0%	1
xilinx	0%	15

Commentary

- Fixing some DTBs check warnings might require testing and many ARM platforms are now legacy
    - Some are still actively developed, like Aspeed, but not much happening in fixing warnings
- Effort is mostly moved to ARM64

Status of ARM64 Platforms

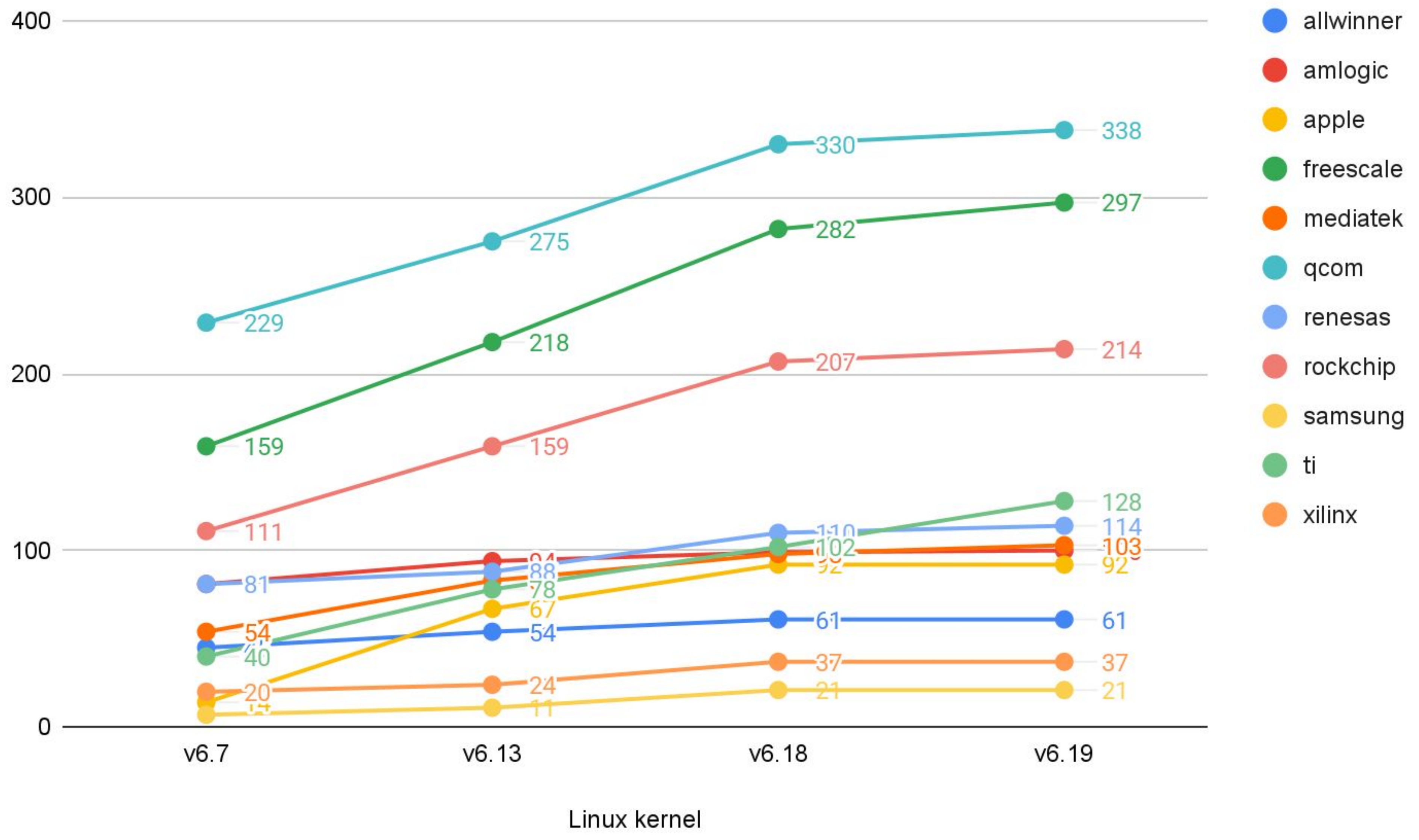
SoC Platforms - ARM64

- There is 40 subdirectories in arch/arm64/boot/dts - ~40 SoC vendors
  - Tesla and Exynos are both Samsung
  - Freescale = NXP (old naming used here)

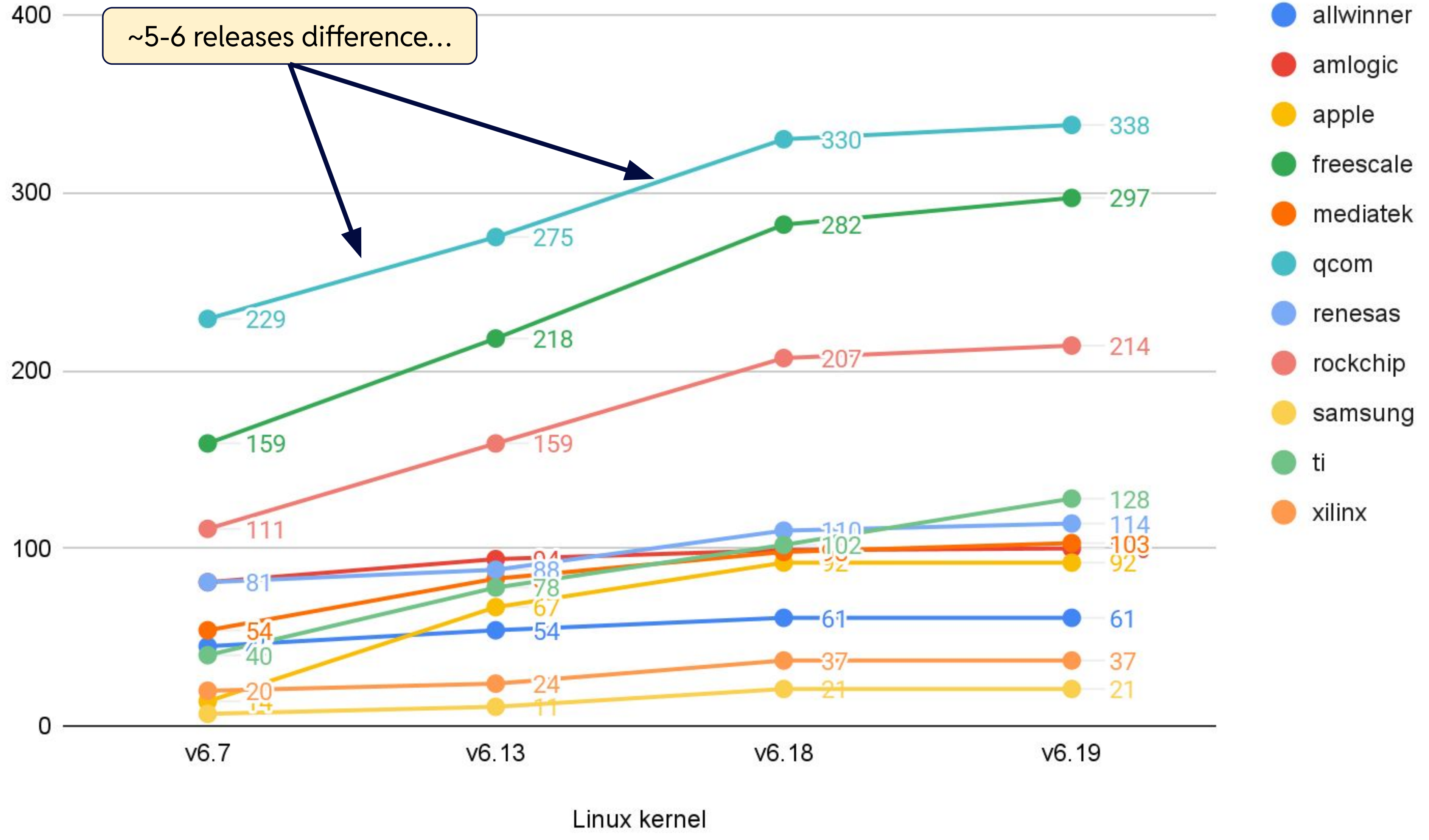
SoC Platforms - ARM64

- There is 40 subdirectories in arch/arm64/boot/dts - ~40 SoC vendors
  - Tesla and Exynos are both Samsung
  - Freescale = NXP (old naming used here)
- Some not being actively developed, at least in DTS
  - They still could be well supported ACPI platform...
- Following slides show subset of architectures, e.g. most active in Linux kernel

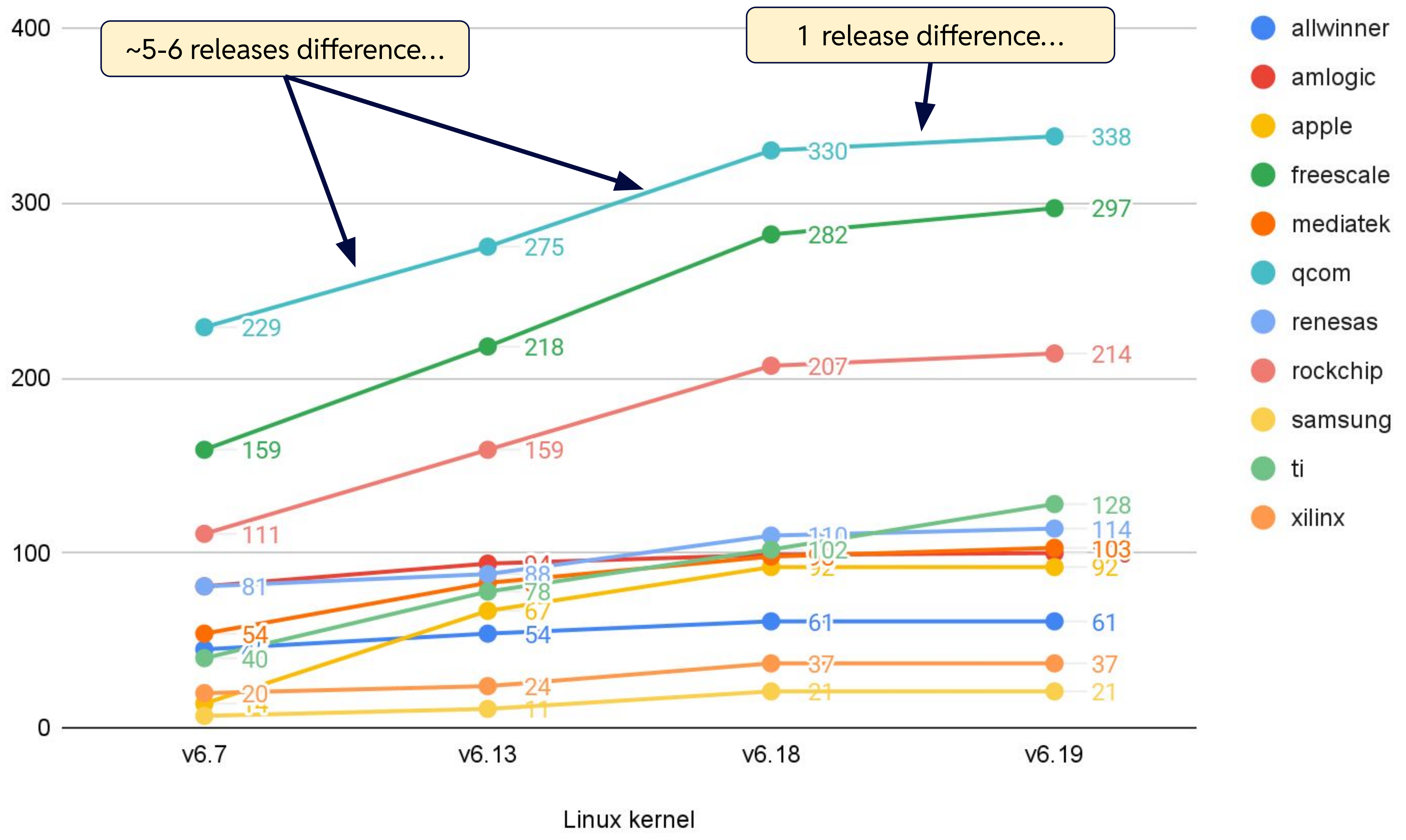
Number of Targets - ARM64 (Most Active)



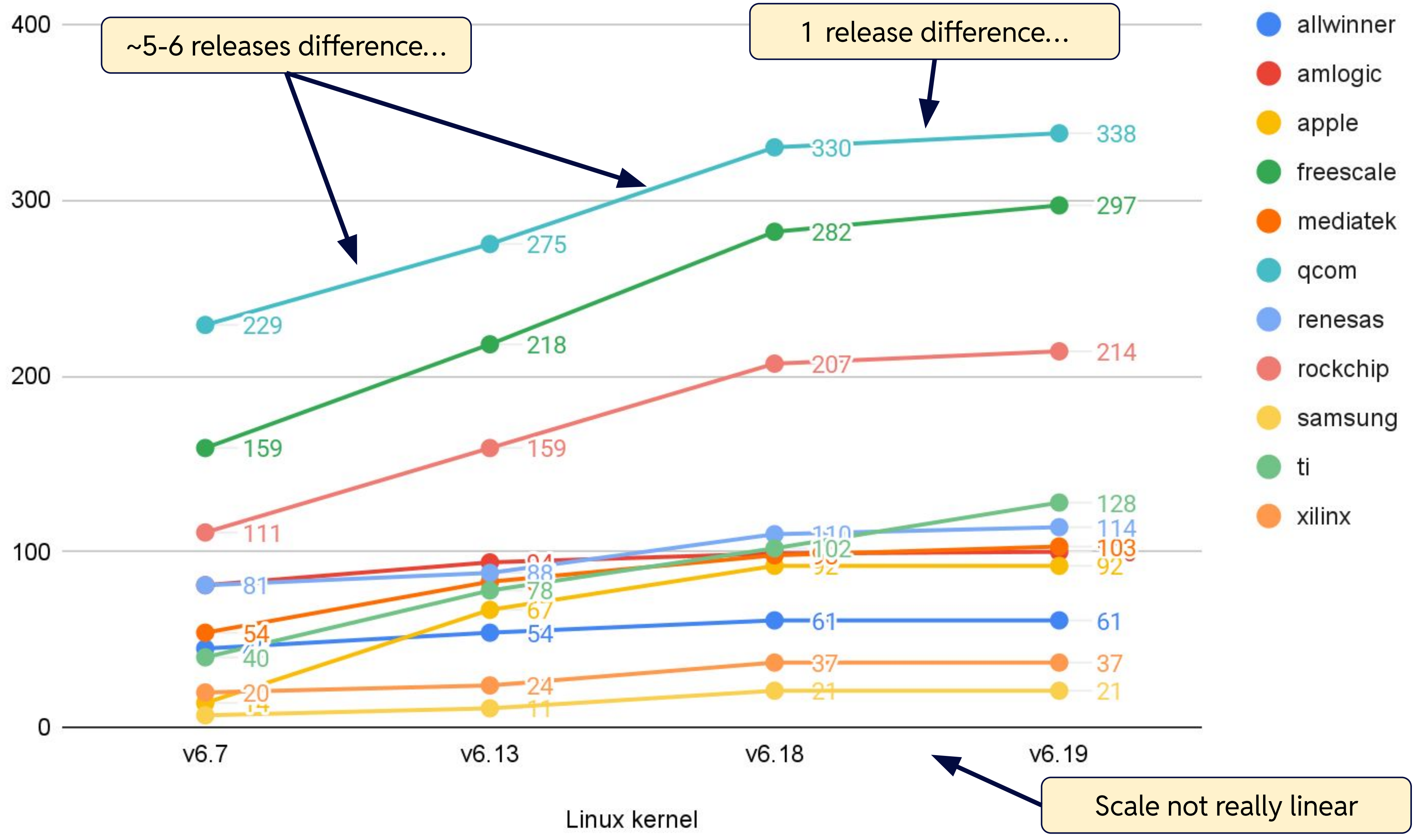
Number of Targets - ARM64 (Most Active)



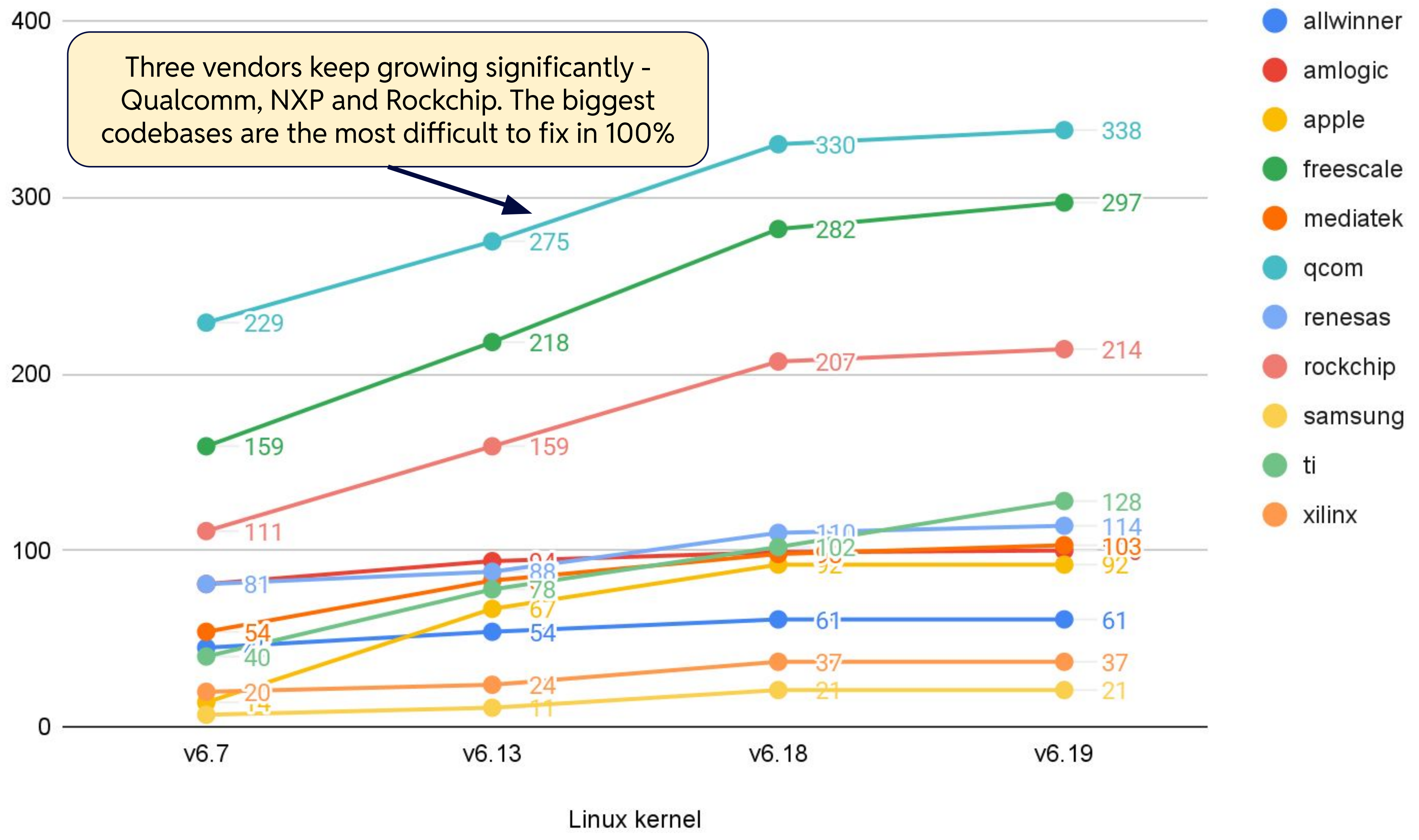
Number of Targets - ARM64 (Most Active)



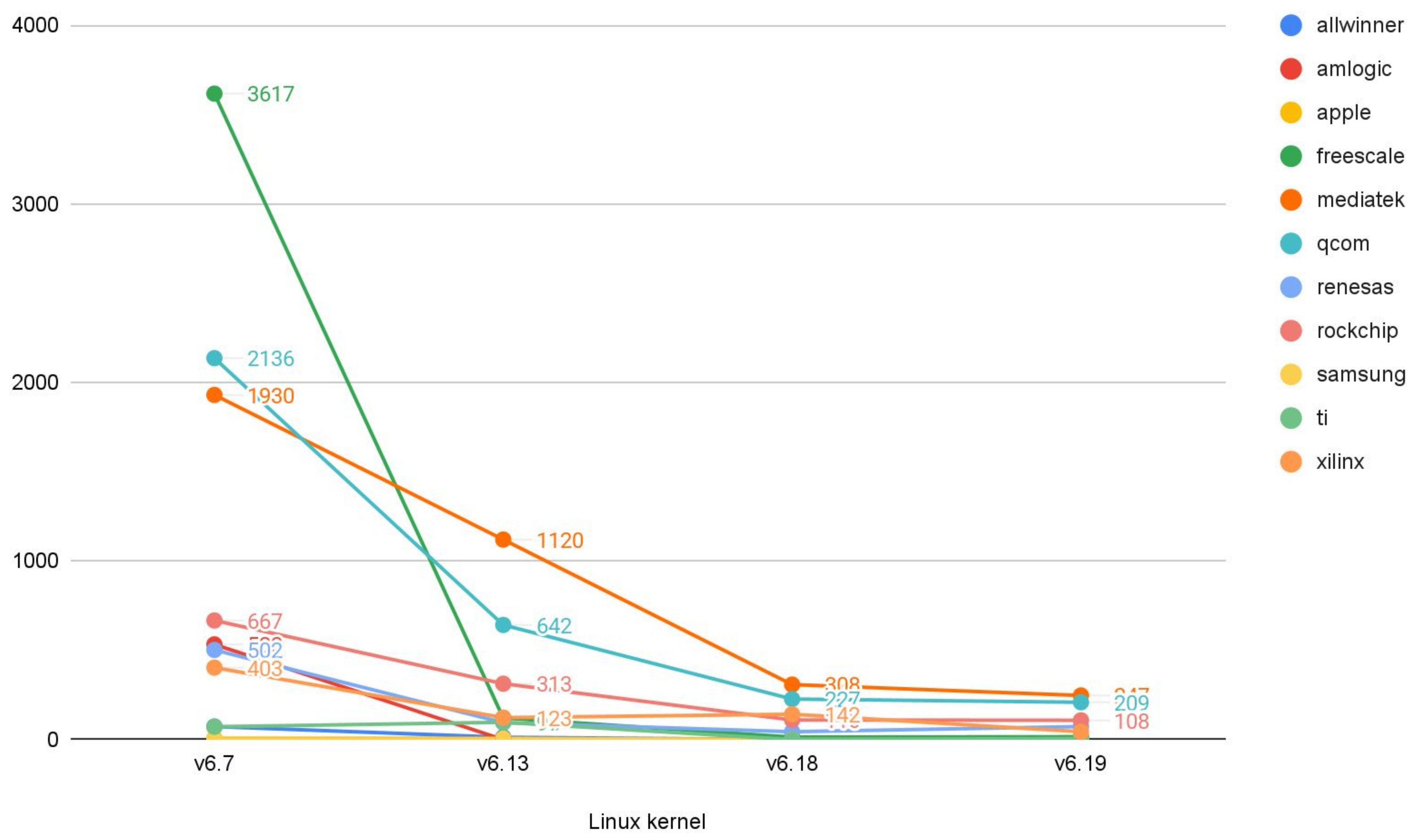
Number of Targets - ARM64 (Most Active)



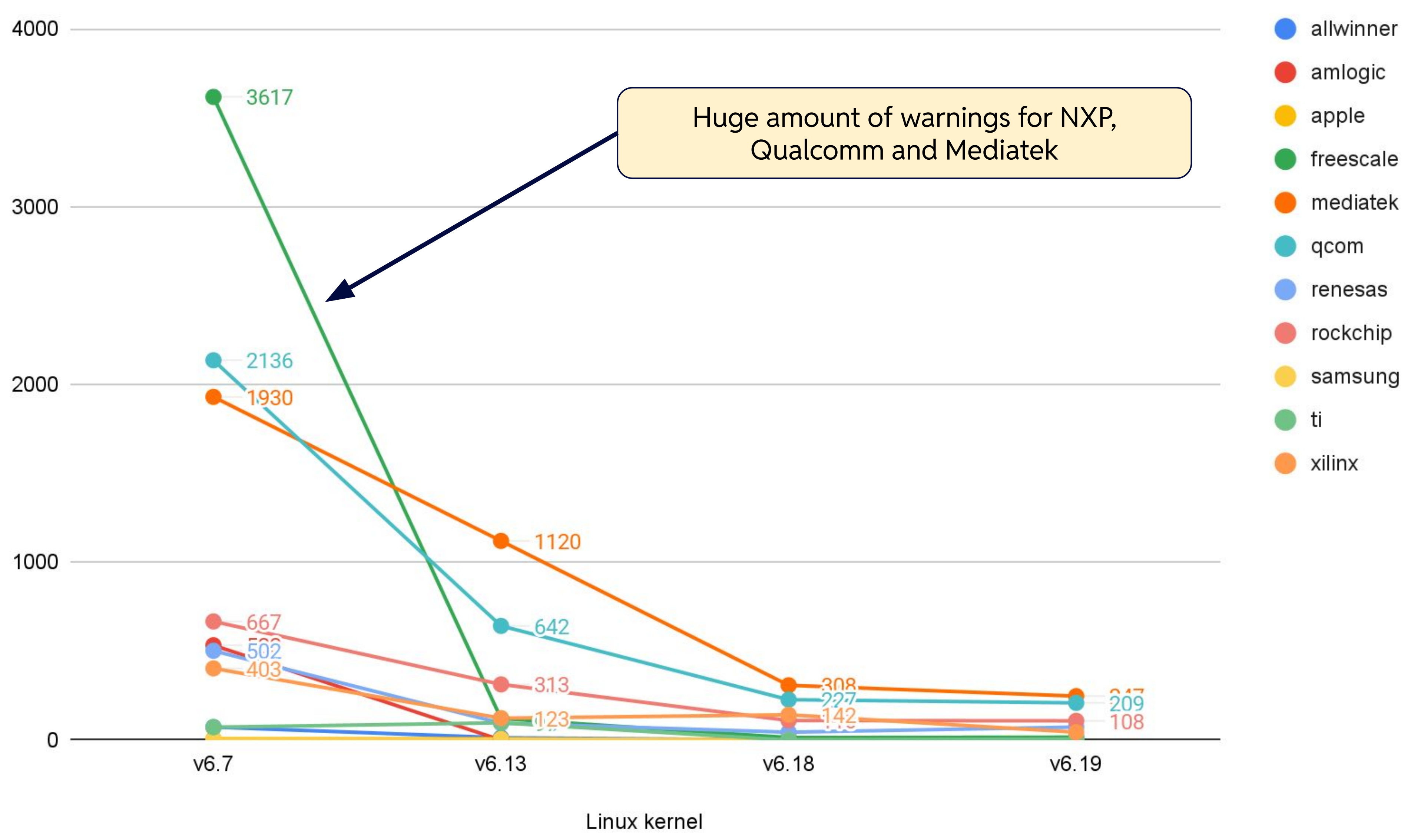
Number of Targets - ARM64 (Most Active)



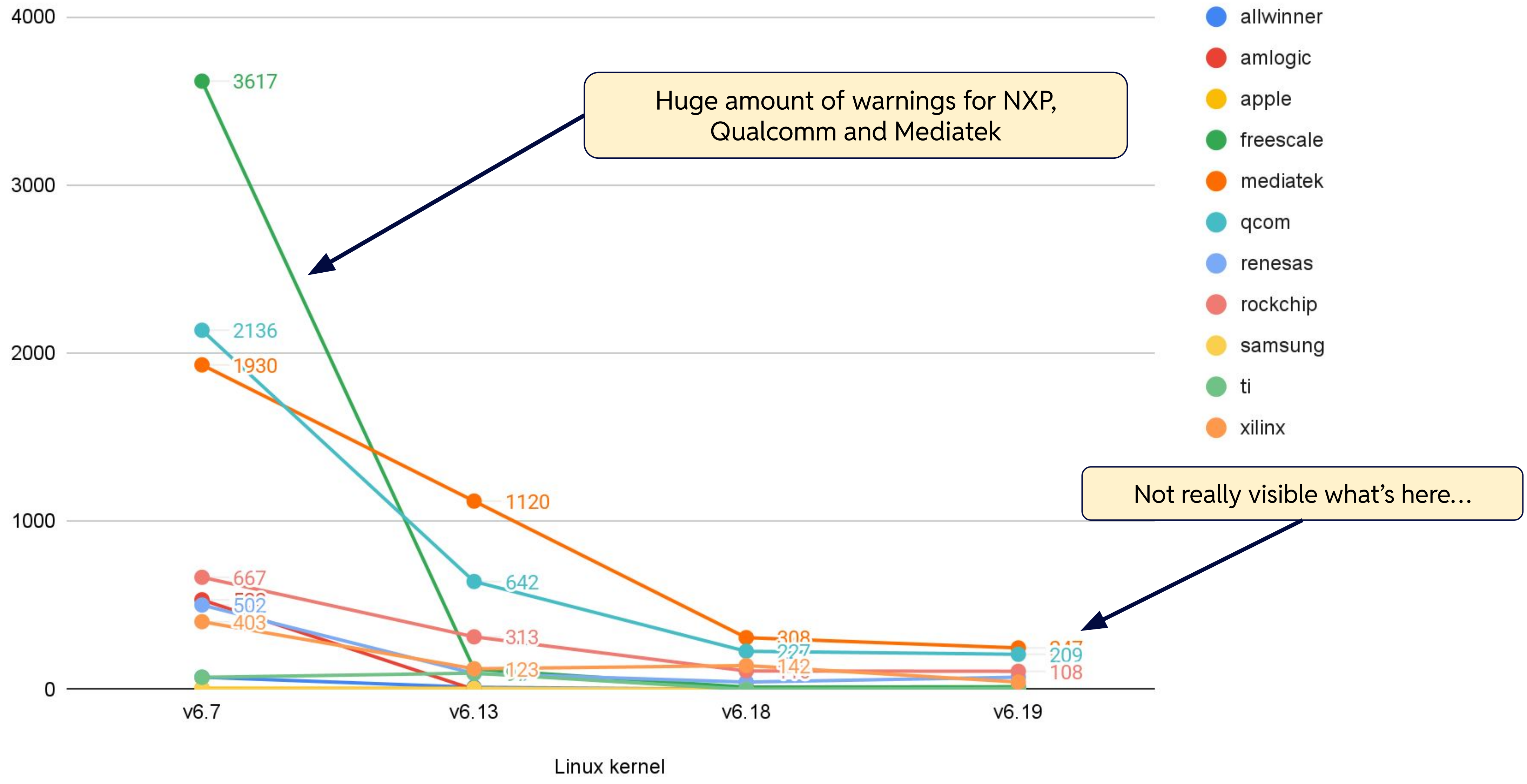
Total DTBs Check Warnings (Lower Better) - ARM64 (Most Active)



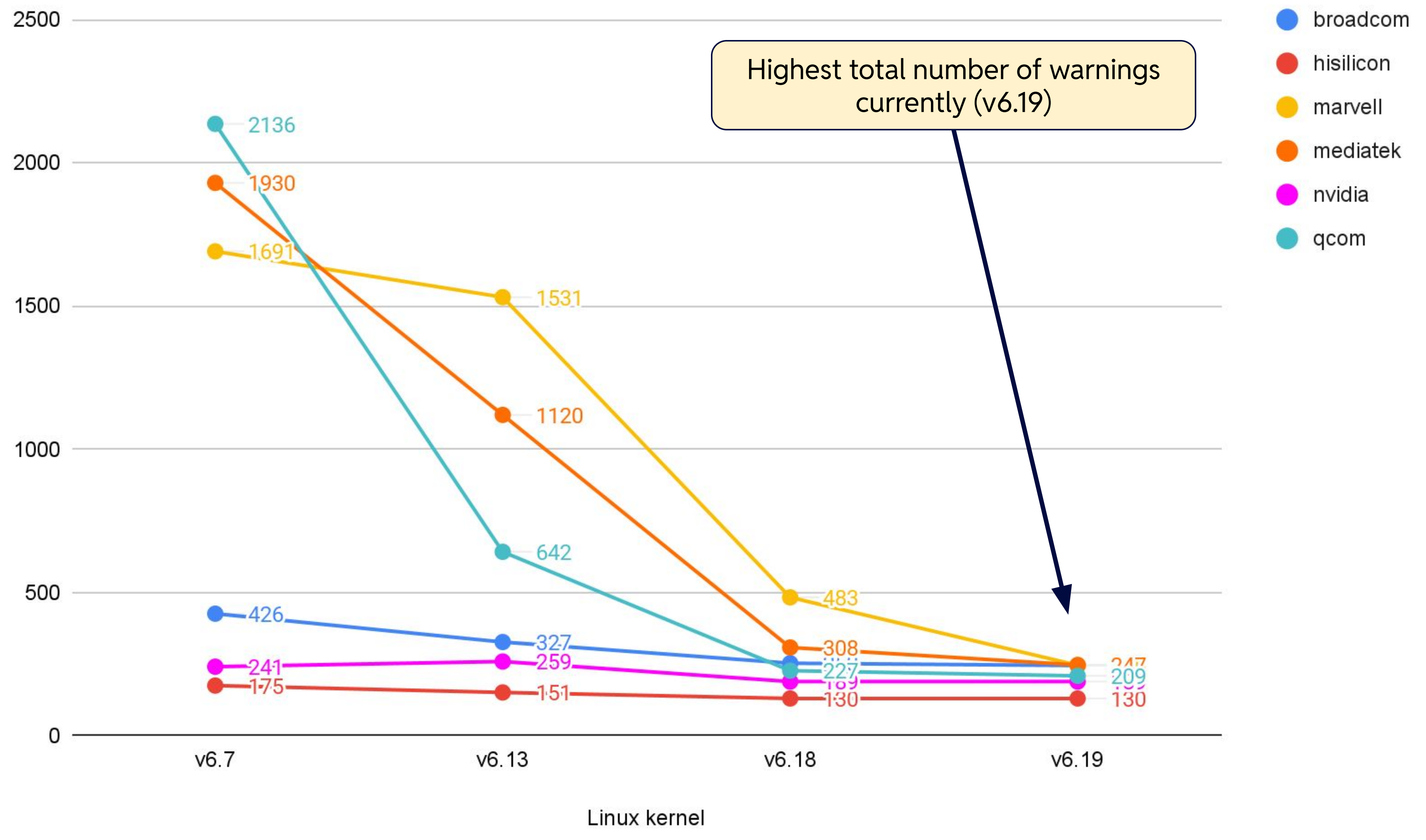
Total DTBs Check Warnings (Lower Better) - ARM64 (Most Active)



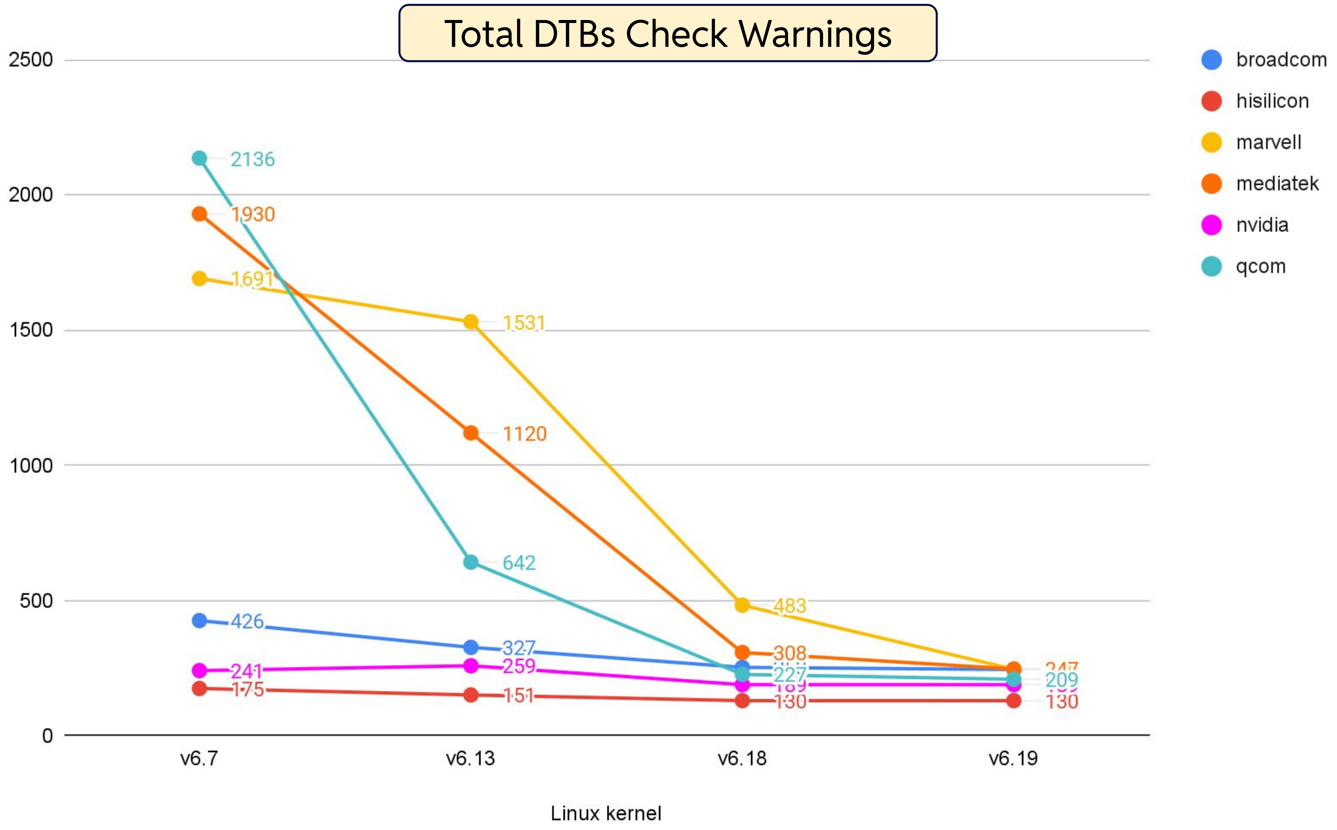
Total DTBs Check Warnings (Lower Better) - ARM64 (Most Active)



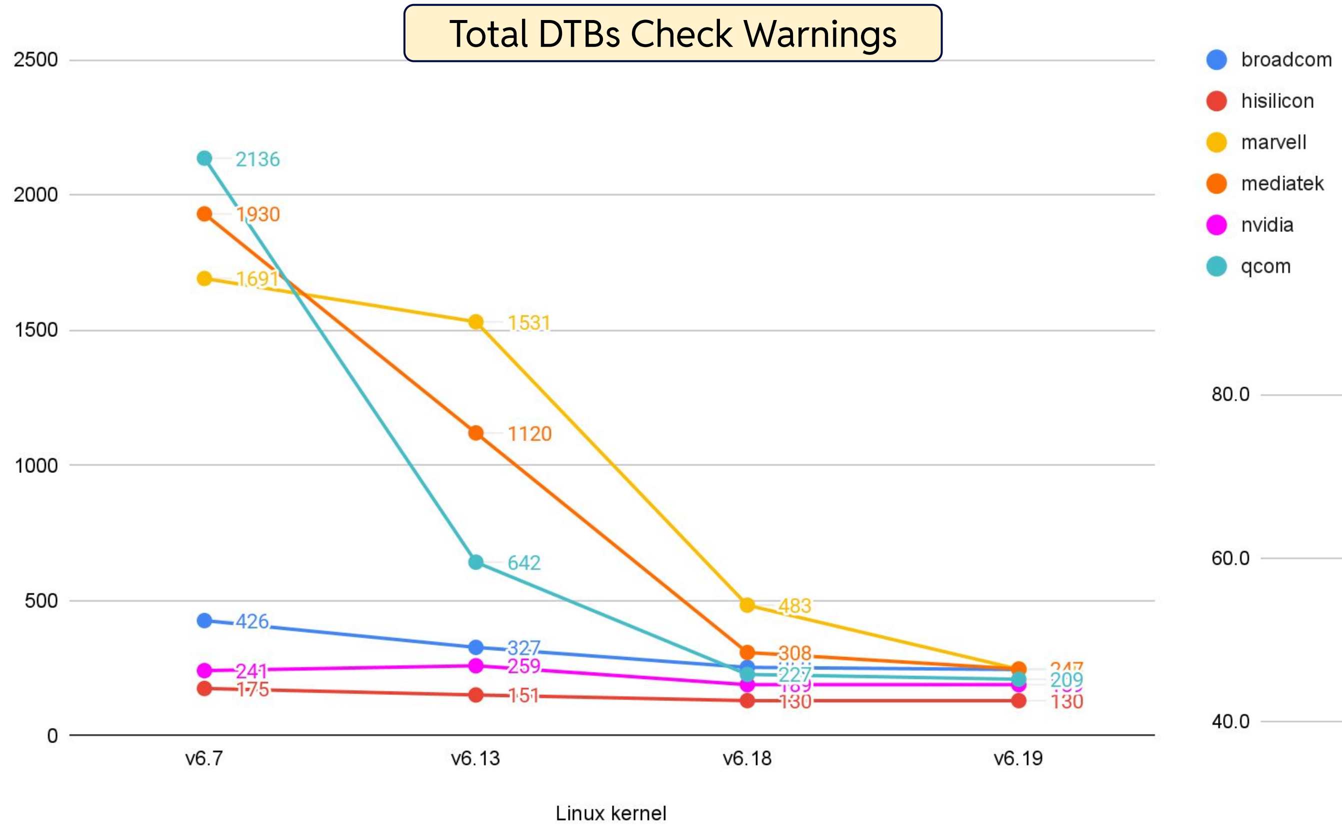
Total DTBs Check Warnings (Lower Better) - Highest To. Numbers

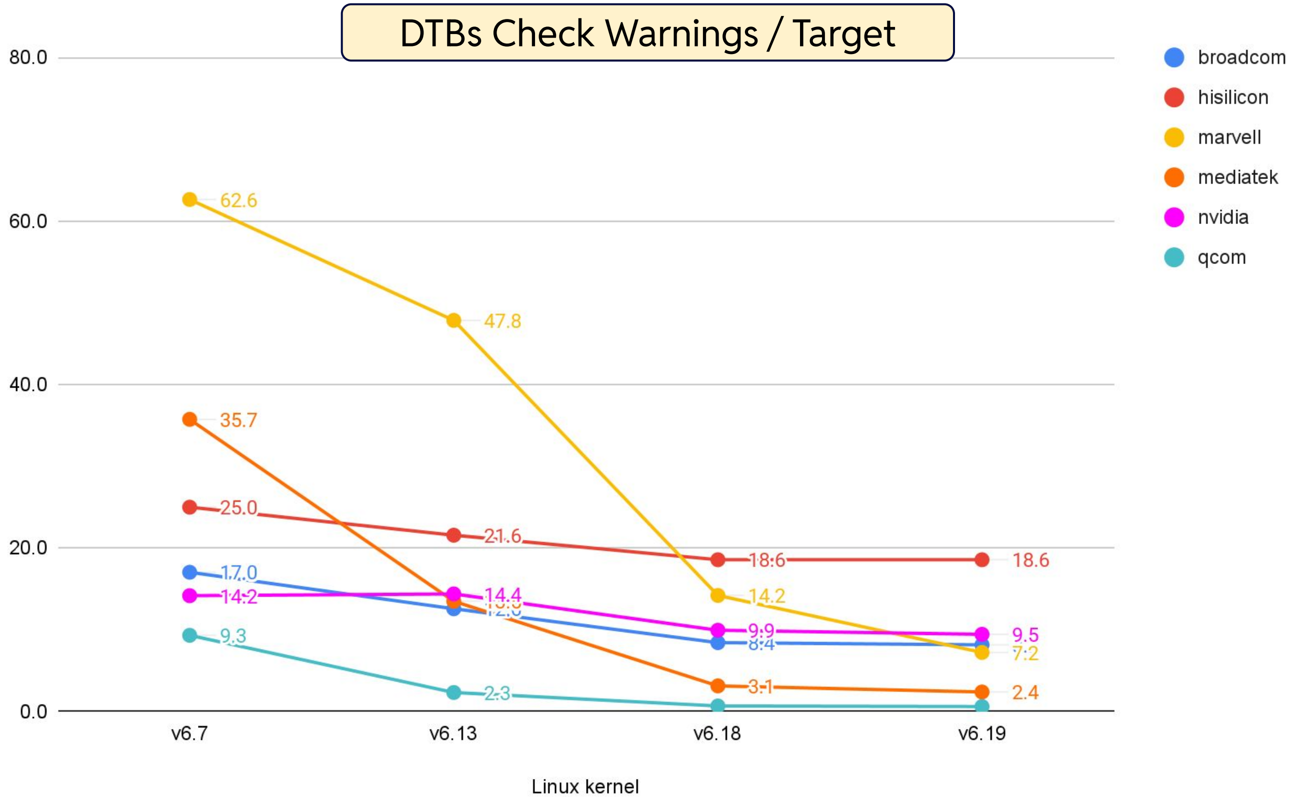


Total and DTBs Check Warnings / Target (Lower Better)

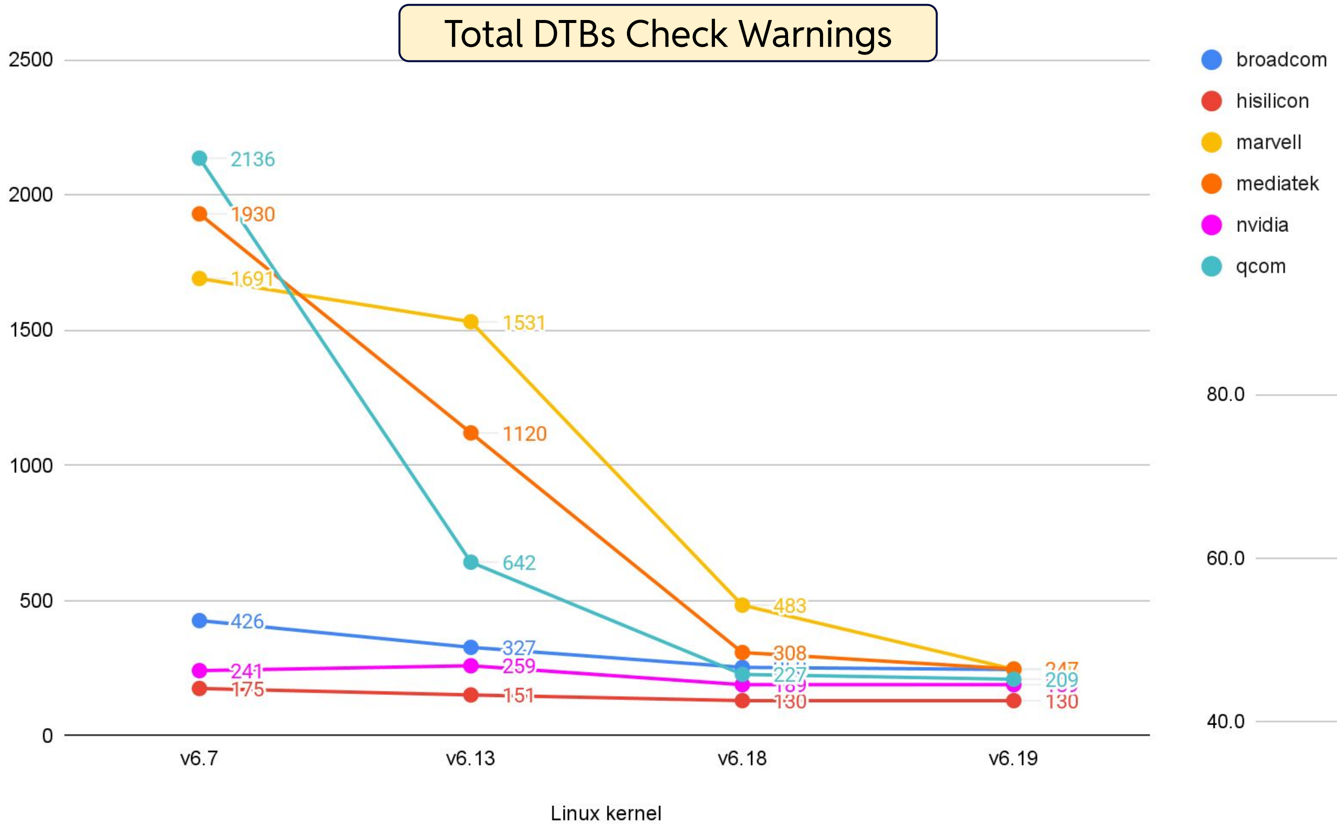


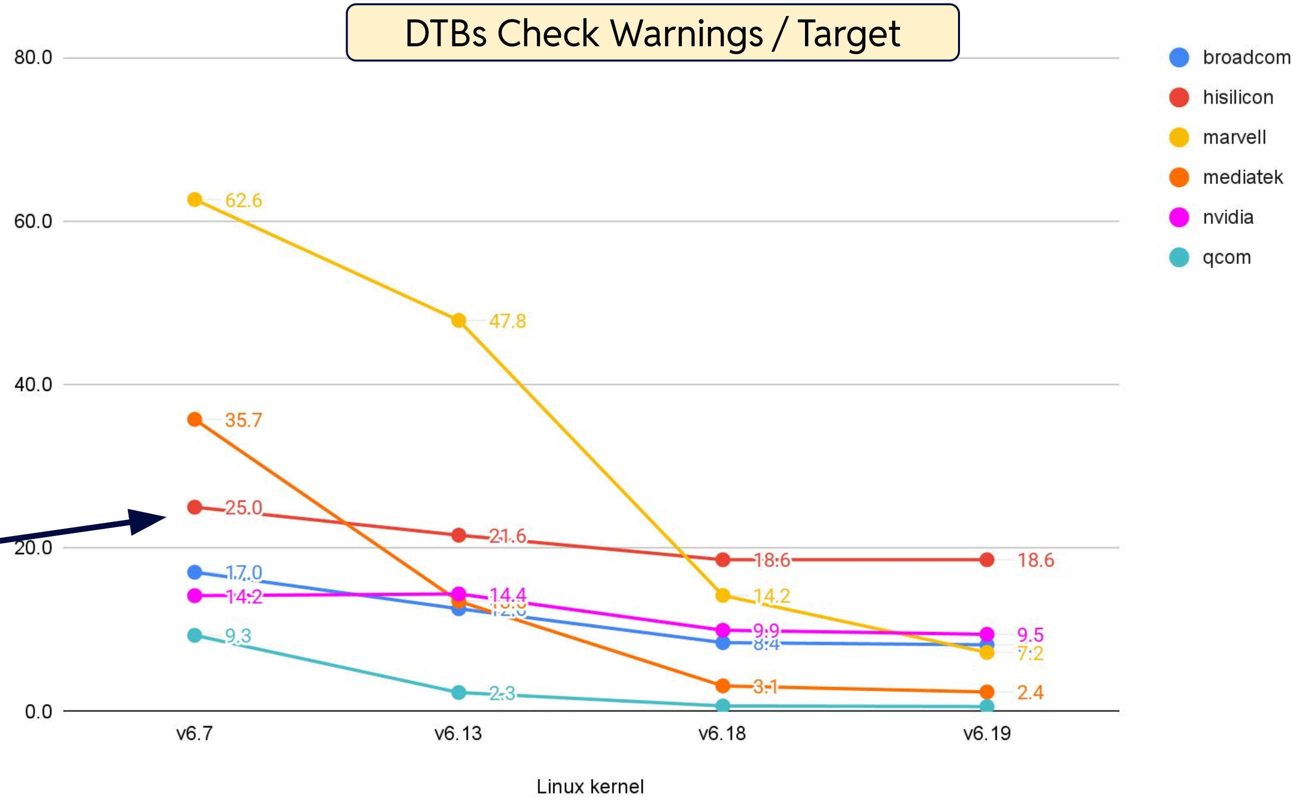
Total and DTBs Check Warnings / Target (Lower Better)





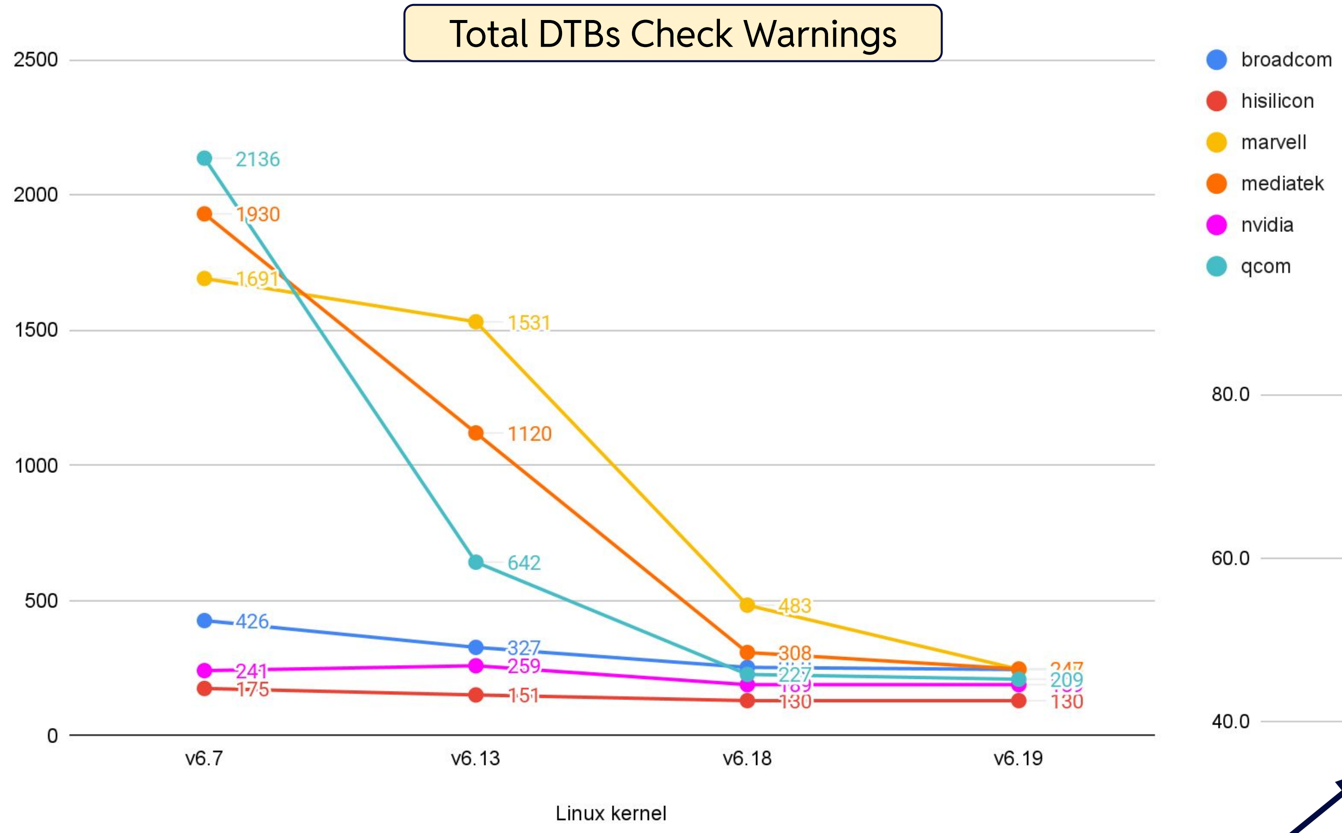
Total and DTBs Check Warnings / Target (Lower Better)

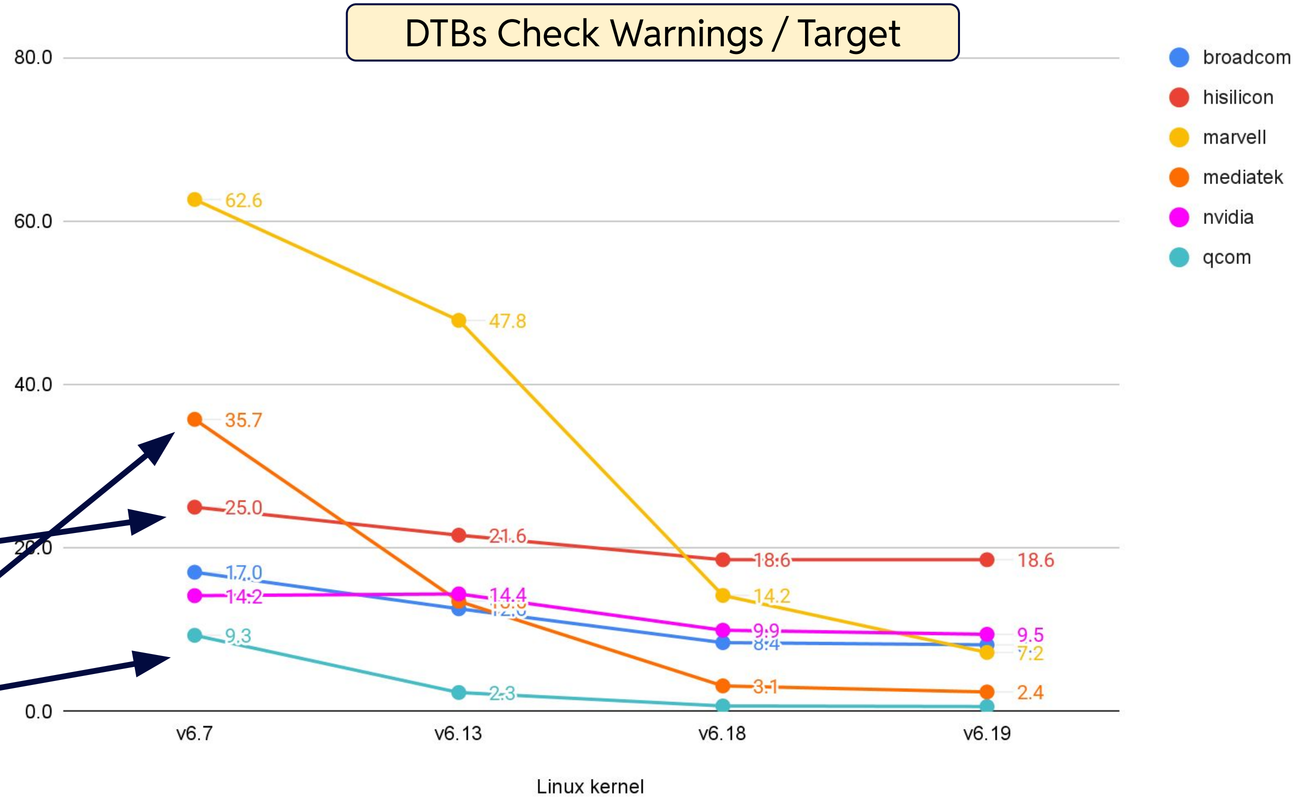




HiSilicon is not an active platform, nothing changing.
Nvidia and Broadcom are moderately active, but not improving

Total and DTBs Check Warnings / Target (Lower Better)

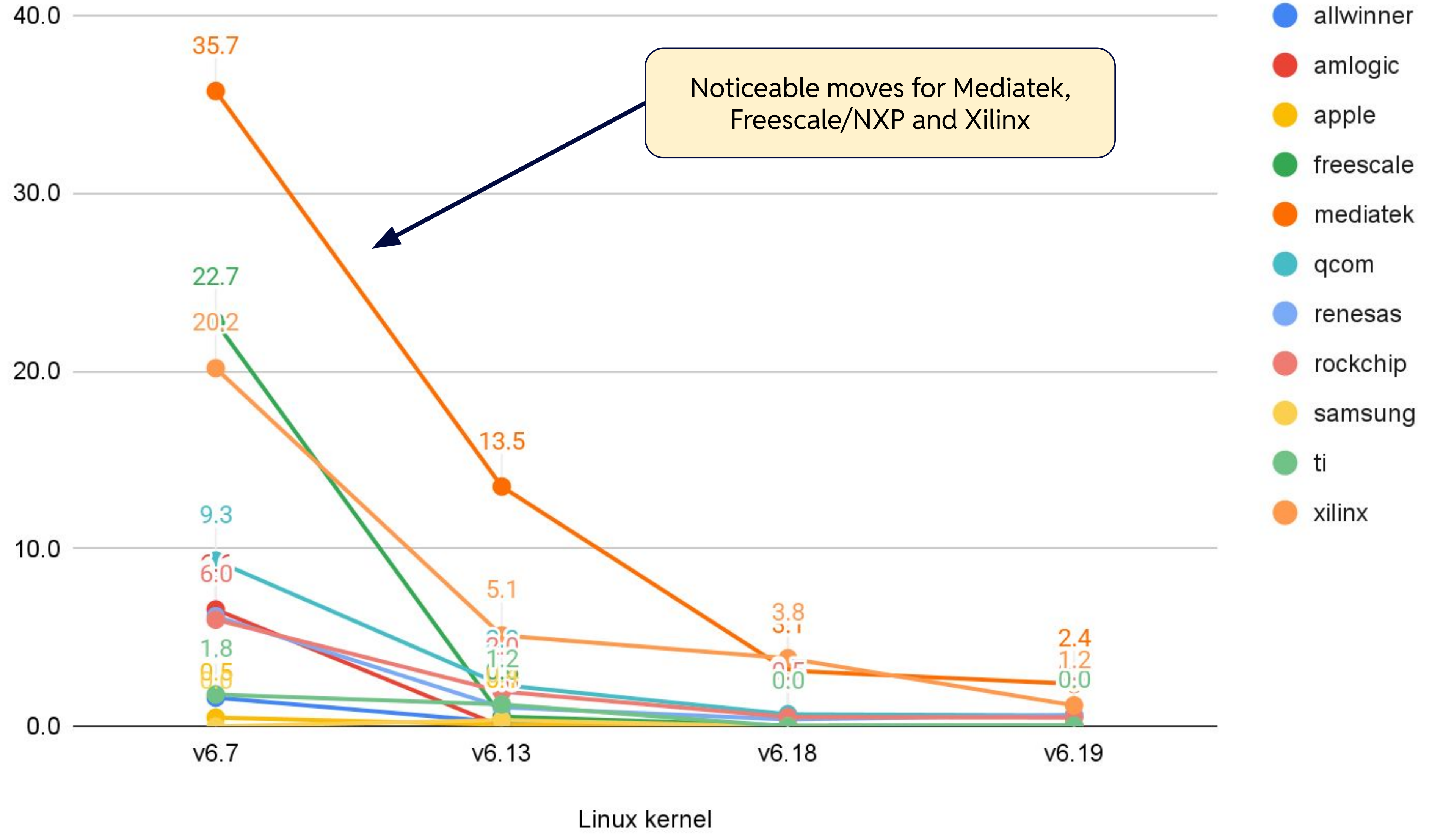




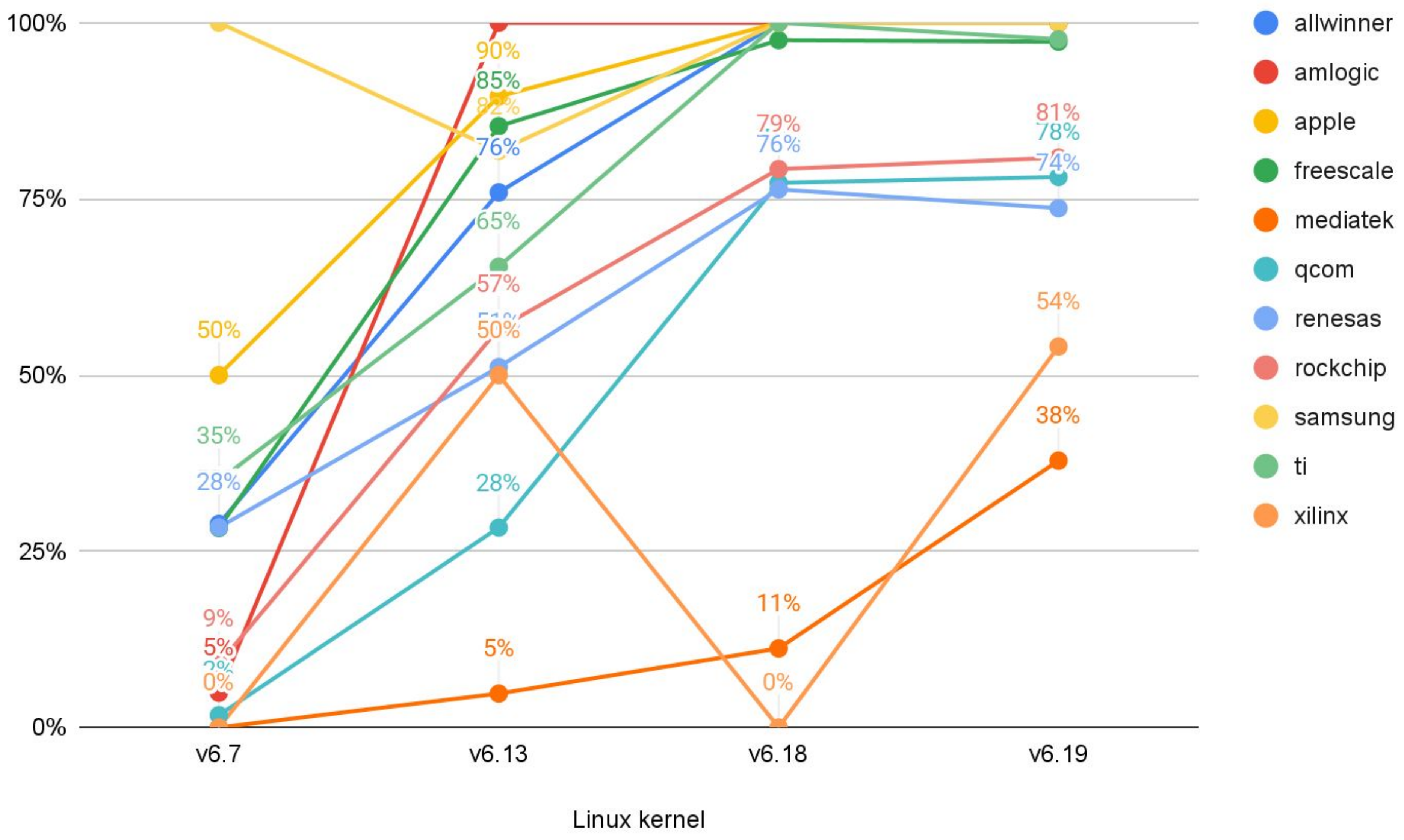
HiSilicon is not an active platform, nothing changing. Nvidia and Broadcom are moderately active, but not improving

Qualcomm and MediaTek, very high in total numbers, are doing actually well

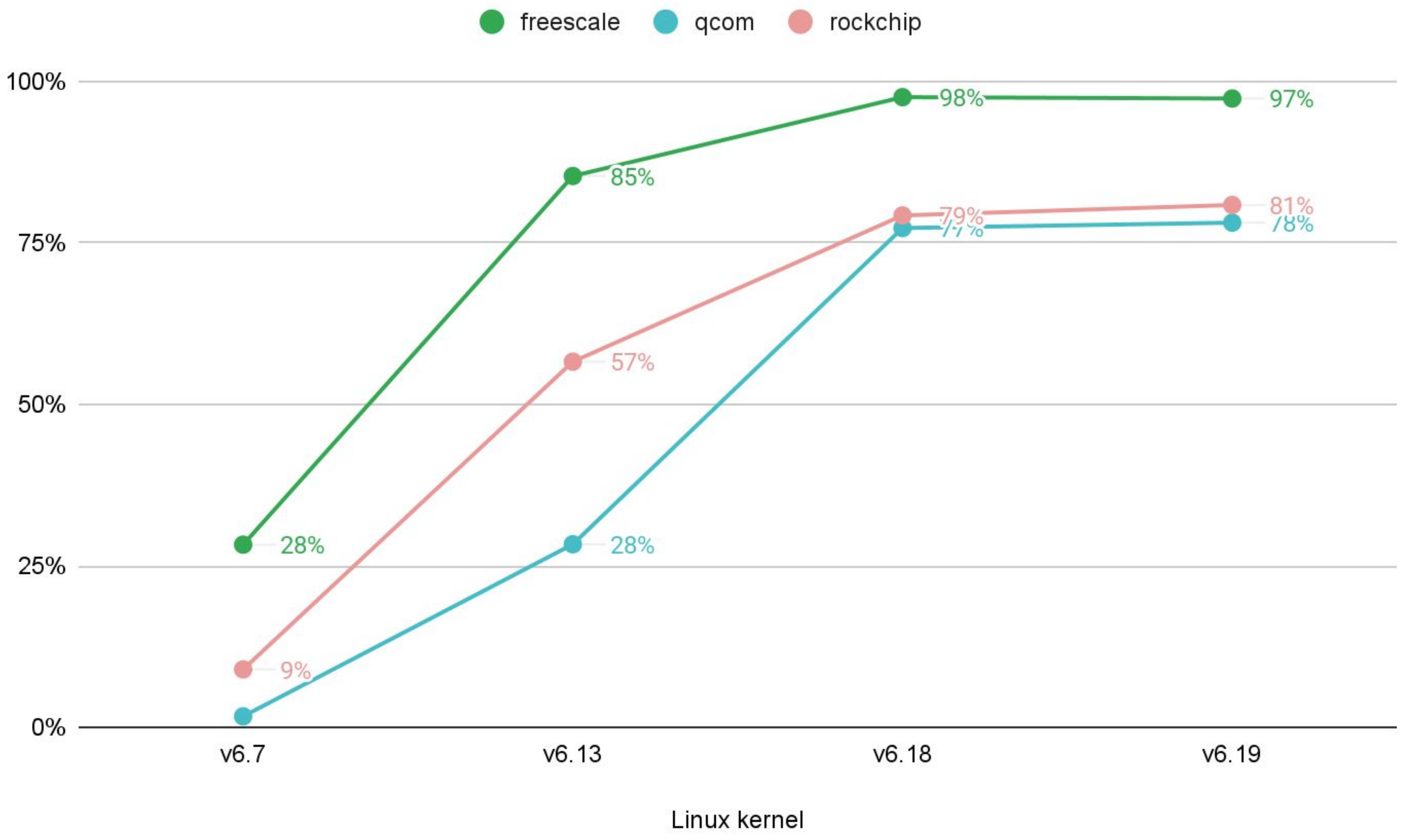
DTBs Check Warnings / Target (Lower Better) - ARM64 (Most Activ



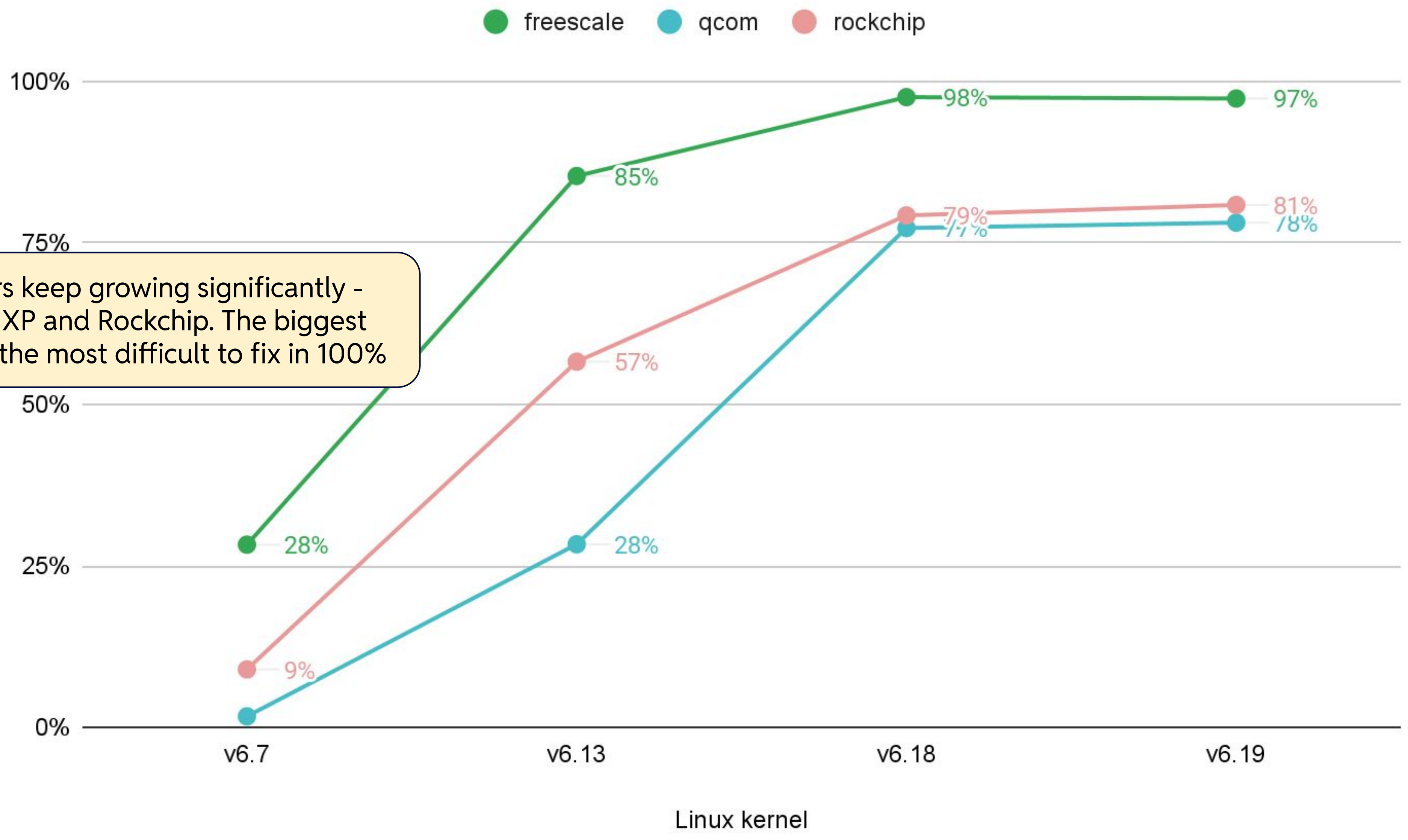
Percentage of Warning-free Targets (Higher Better) - ARM64



Percentage of Warning-free Targets (Higher Better) - ARM64

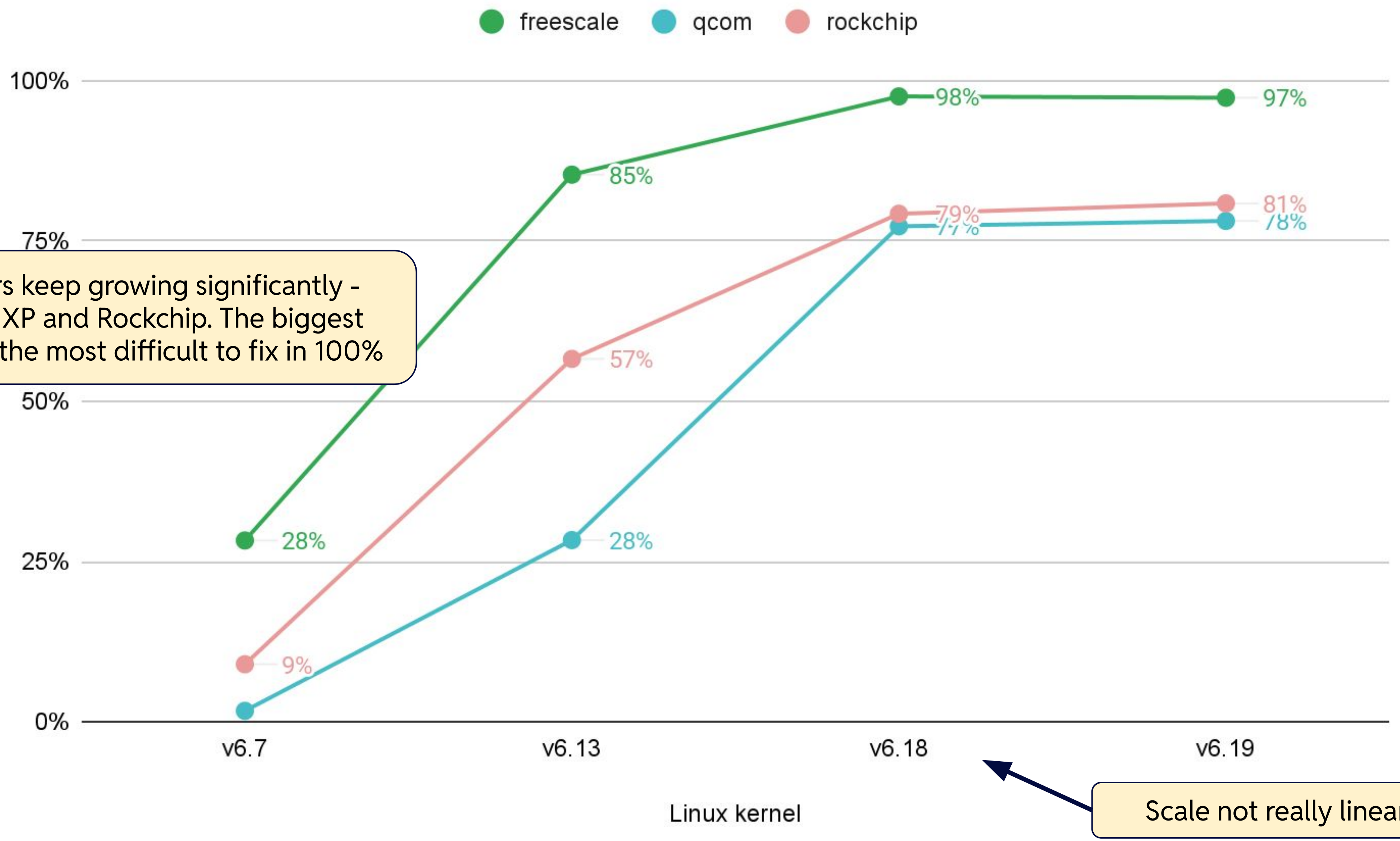


Percentage of Warning-free Targets (Higher Better) - ARM64



Three vendors keep growing significantly - Qualcomm, NXP and Rockchip. The biggest codebases are the most difficult to fix in 100%

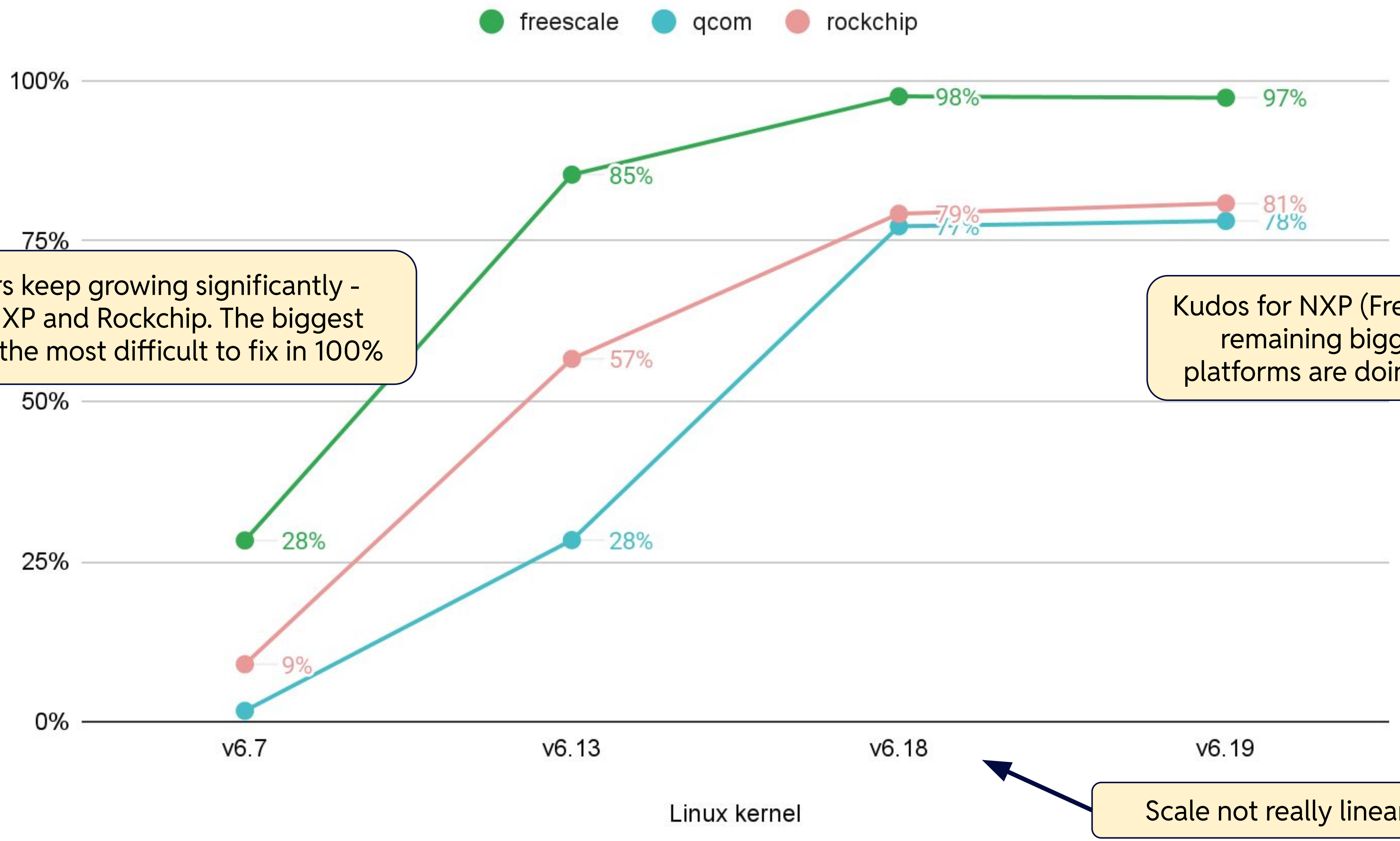
Percentage of Warning-free Targets (Higher Better) - ARM64



Three vendors keep growing significantly - Qualcomm, NXP and Rockchip. The biggest codebases are the most difficult to fix in 100%

Scale not really linear

Percentage of Warning-free Targets (Higher Better) - ARM64

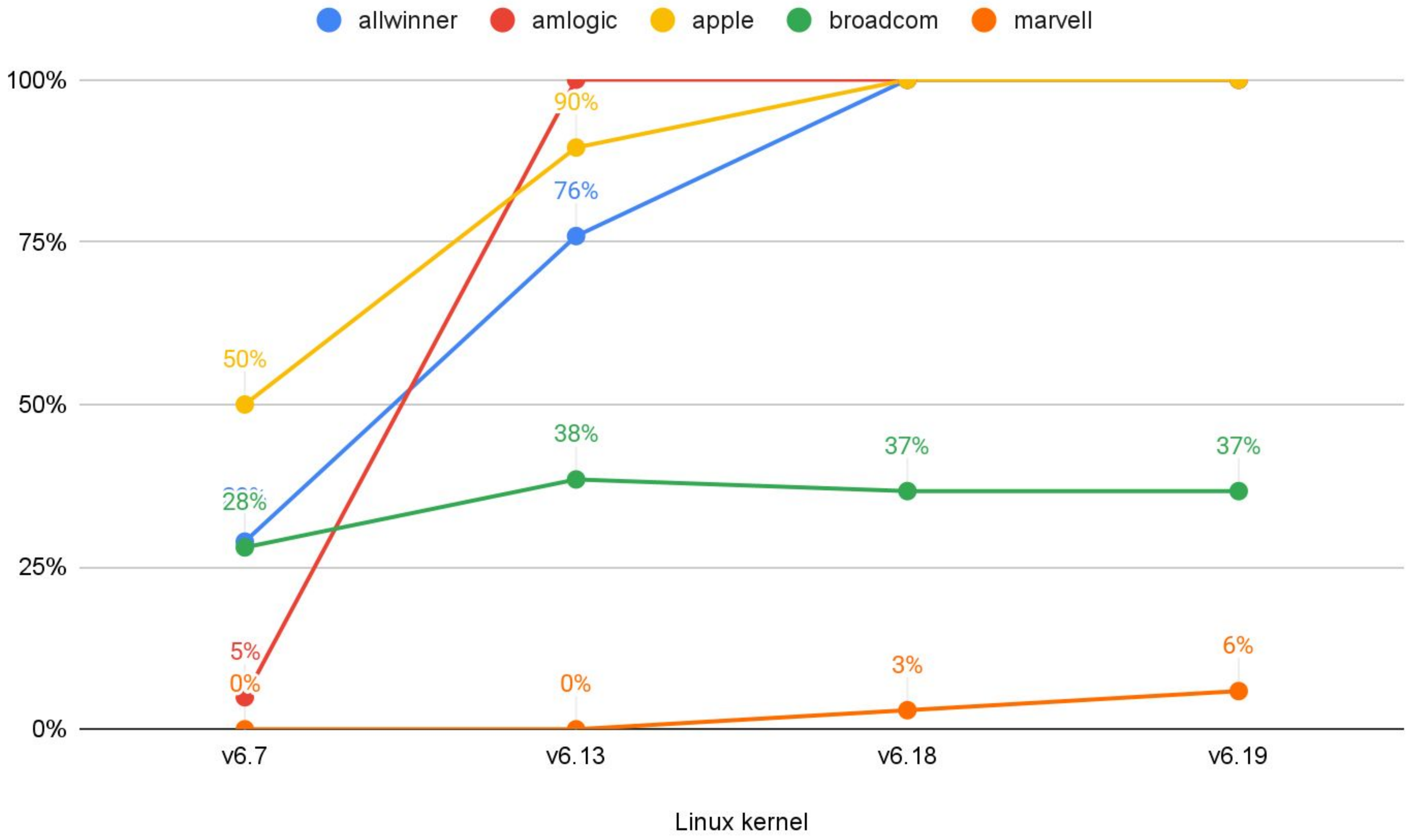


Three vendors keep growing significantly - Qualcomm, NXP and Rockchip. The biggest codebases are the most difficult to fix in 100%

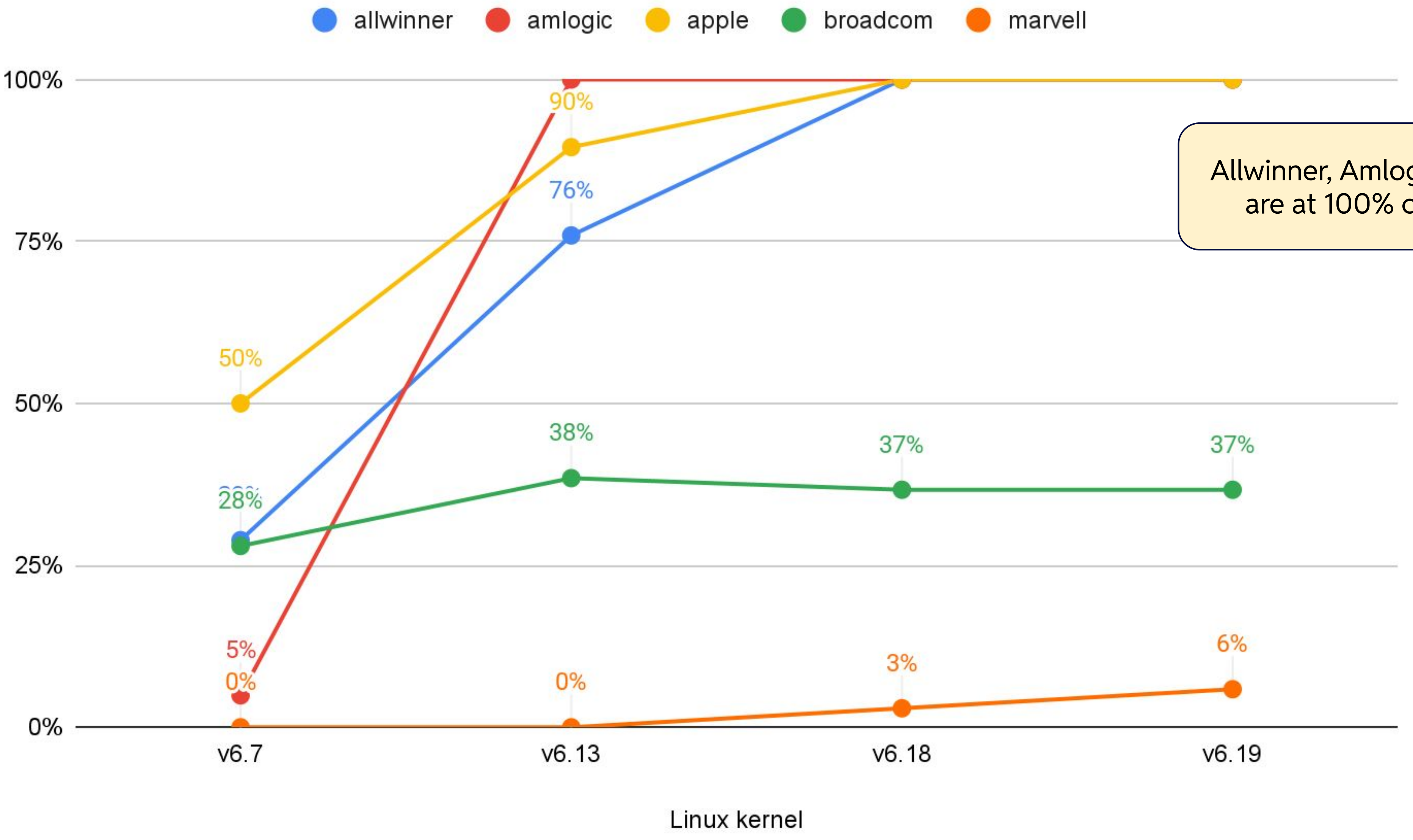
Kudos for NXP (Freescale). Two remaining biggest SoC platforms are doing well too.

Scale not really linear

Percentage of Warning-free Targets (Higher Better) - ARM64

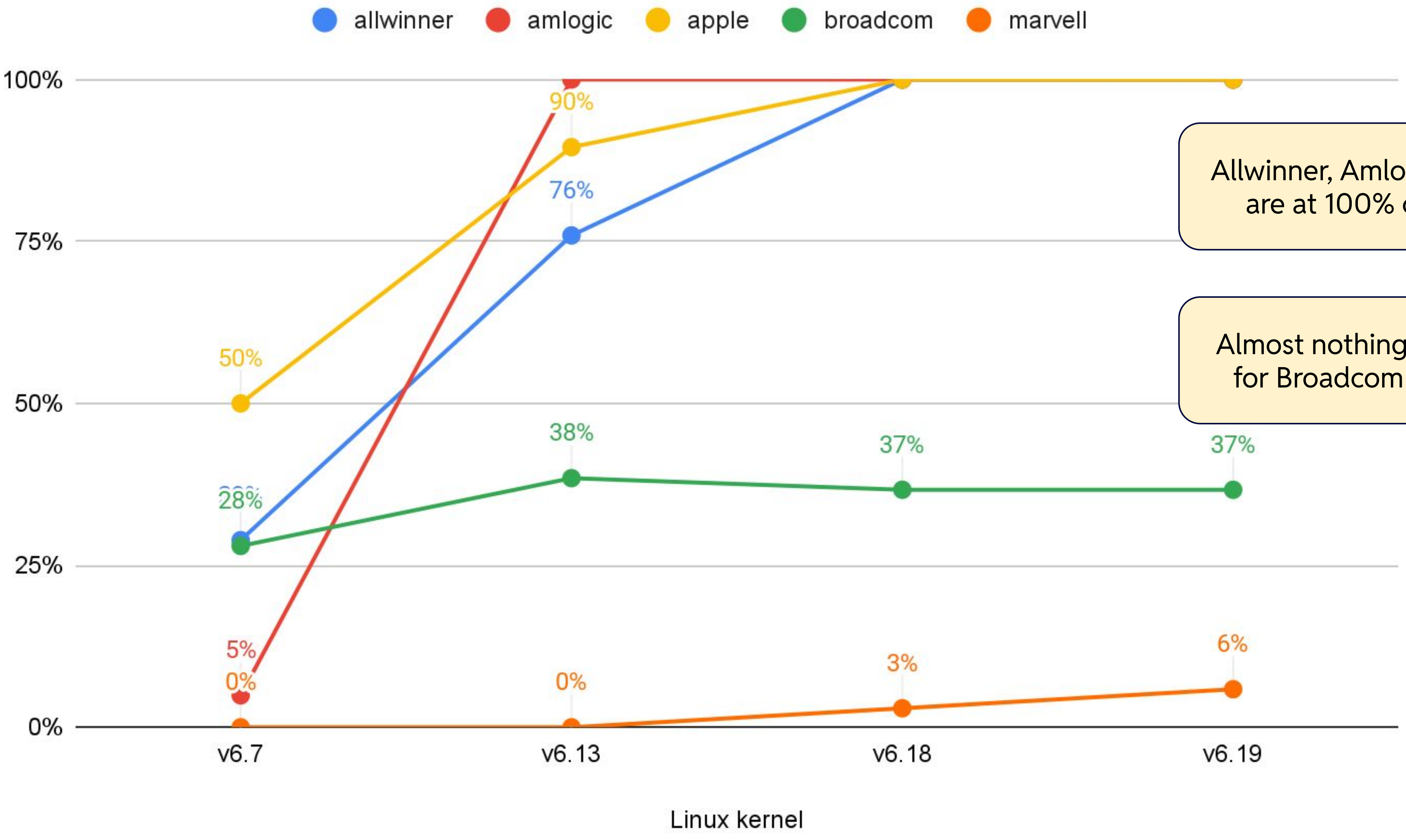


Percentage of Warning-free Targets (Higher Better) - ARM64



Allwinner, Amlogic and Apple are at 100% compliance

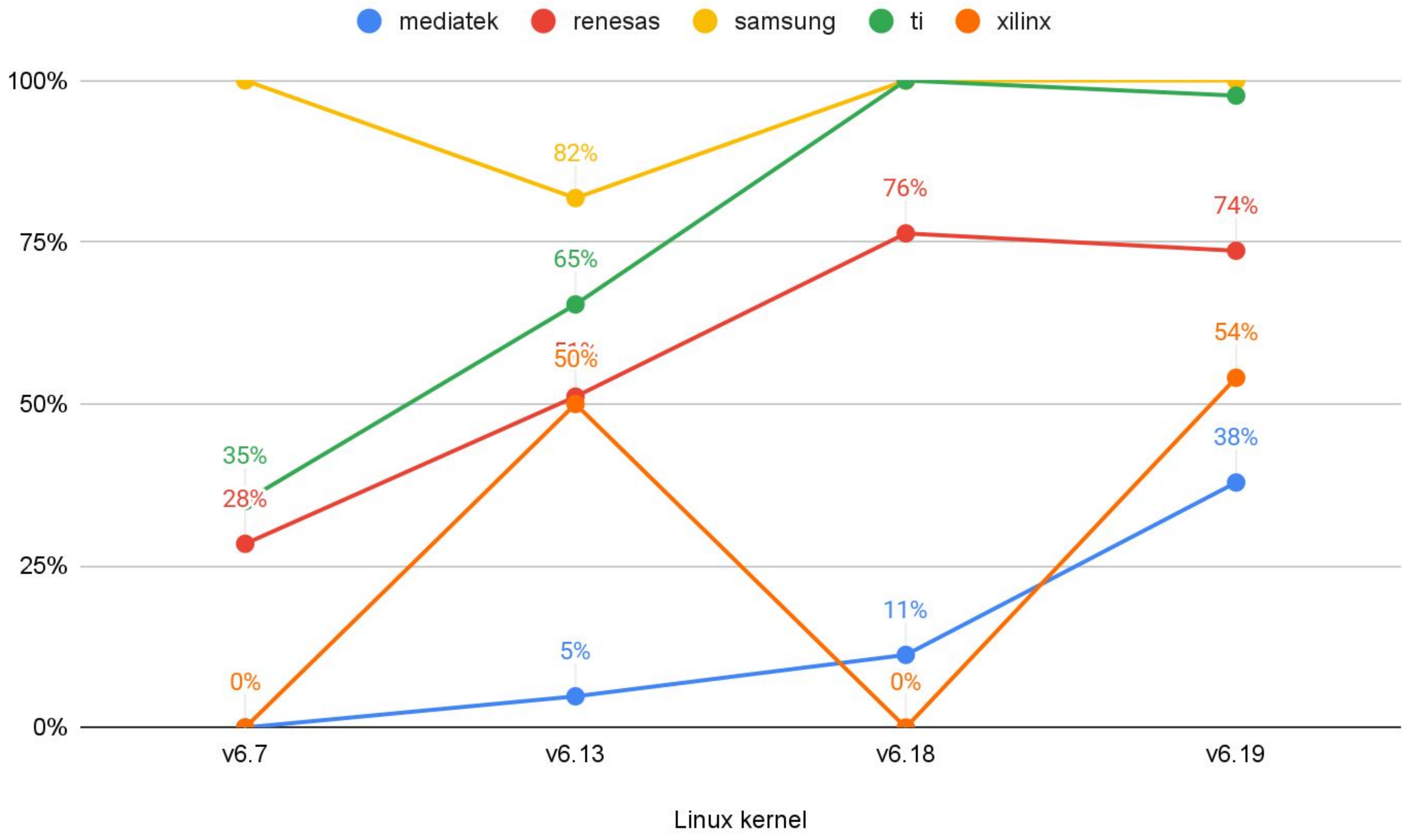
Percentage of Warning-free Targets (Higher Better) - ARM64



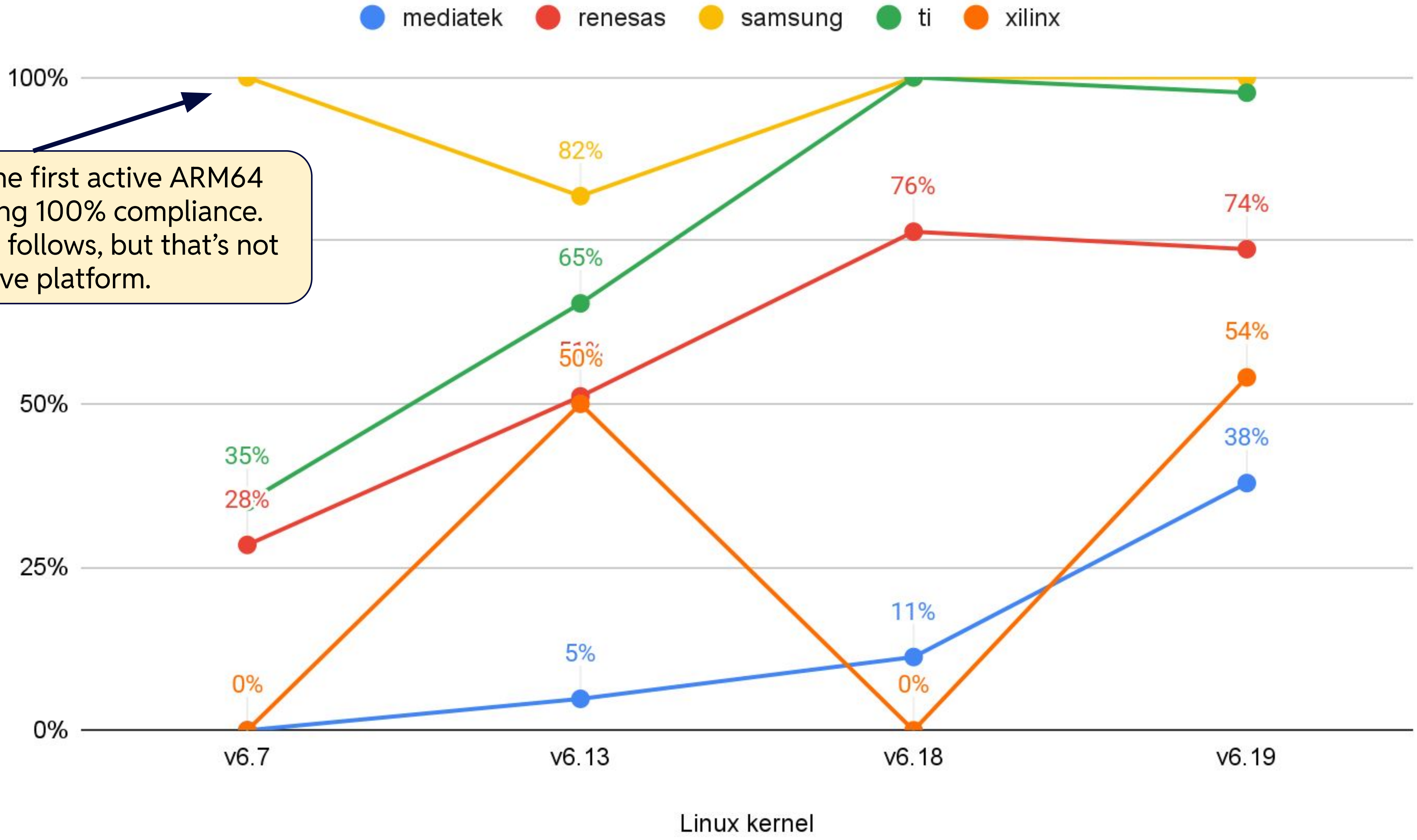
Allwinner, Amlogic and Apple are at 100% compliance

Almost nothing is happening for Broadcom and Marvell

Percentage of Warning-free Targets (Higher Better) - ARM64

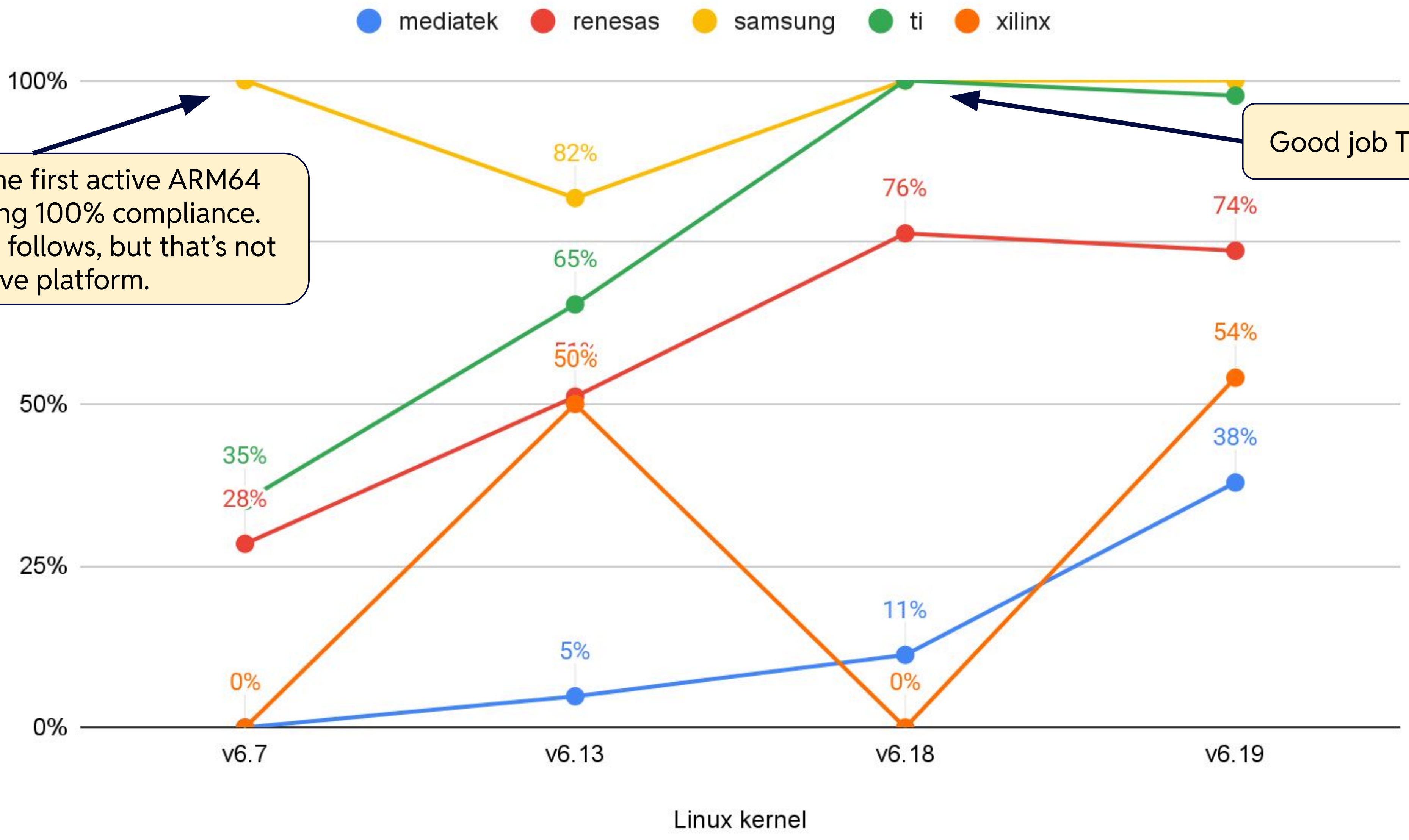


Percentage of Warning-free Targets (Higher Better) - ARM64



Samsung was the first active ARM64 platform reaching 100% compliance. ST (with 1 target) follows, but that's not an active platform.

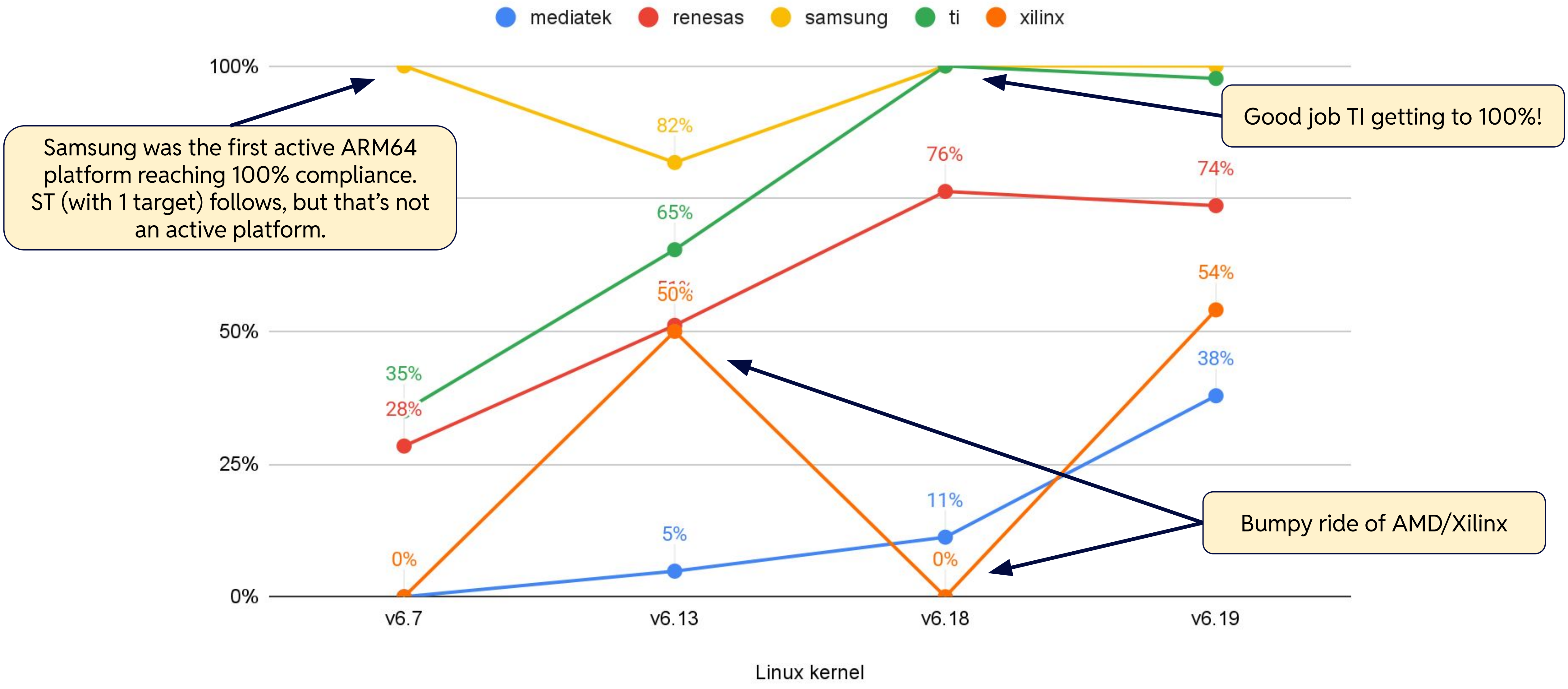
Percentage of Warning-free Targets (Higher Better) - ARM64



Samsung was the first active ARM64 platform reaching 100% compliance. ST (with 1 target) follows, but that's not an active platform.

Good job TI getting to 100%!

Percentage of Warning-free Targets (Higher Better) - ARM64



Warning-free Platforms (Higher Better) - ARM64, v6.19 (next)

SoC Platform	Percentage of Warning-free Targets	No. Targets	LoC
actions	100%	2	1007
airoha	100%	1	509
allwinner	100%	61	19096
altera	100%	3	1158
amazon	100%	2	725
amlogic	100%	100	37138
apple	100%	92	26375
axiado	100%	1	604
blaize	100%	1	361

SoC Platform	Percentage of Warning-free Targets	No. Targets	LoC
bst	100%	1	123
cix	100%	1	1080
intel	100%	9	2718
samsung	100%	21	36021
sophgo	100%	1	204
synaptics	100%	2	378
ti	97.6%	128	69486
freescale	97.3%	297	155467

Platforms with Only Warnings - ARM64, v6.19 (next)

SoC Platform	Percentage of Warning-free Targets	No. Targets	LoC
apm	0%	2	2094
bitmain	0%	1	413
cavium	0%	2	660
hisilicon	0%	7	12743
lg	0%	2	478
microchip	0%	5	2403
nvidia	0%	20	45495
realtek	0%	9	1203
socionext	0%	8	3580
toshiba	0%	2	788

Commentary

- Samsung was the first active ARM64 platform reaching 100% compliance.

Commentary

- Samsung was the first active ARM64 platform reaching 100% compliance.
    - Took me a lot of time to get there

Commentary

- Samsung was the first active ARM64 platform reaching 100% compliance.
  - Took me a lot of time to get there
  - Would not be able to keep it at 100% without automated testing / CI
    - Thank you contributors for keeping it at 100% :)

Commentary

- Samsung was the first active ARM64 platform reaching 100% compliance.
  - Took me a lot of time to get there
  - Would not be able to keep it at 100% without automated testing / CI
    - Thank you contributors for keeping it at 100% :)
- Kudos to several active platforms reaching 100%

Commentary

- Samsung was the first active ARM64 platform reaching 100% compliance.
    - Took me a lot of time to get there
    - Would not be able to keep it at 100% without automated testing / CI
        - Thank you contributors for keeping it at 100% :)
- Kudos to several active platforms reaching 100%
    - Allwinner
    - Amlogic (early, v6.12!)
    - Apple
    - Intel/Altera (at v6.19)
    - NXP/Freescale
    - Texas Instruments

Summary

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
    - RISC-V

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
    - RISC-V
- Is there an architecture with all targets failing (nothing warning-free)?

Quiz - Architectures

- Is there an architecture in the kernel fully compliant (warning-free)?
  - Yes!
    - RISC-V
- Is there an architecture with all targets failing (nothing warning-free)?
  - Yes!
    - PowerPC

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one...

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
    - Amlogic
    - Actions and Airoha (but these are small platforms)

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one...
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
    - Amlogic
    - Actions and Airoha (but these are small platforms)
- Is there an incomplete platform with a very small DTS and still not compliant at all?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
    - Amlogic
    - Actions and Airoha (but these are small platforms)
- Is there an incomplete platform with a very small DTS and still not compliant at all?
  - Yes!
    - More than one…

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
    - Yes!
        - More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
    - Yes!
        - Amlogic
        - Actions and Airoha (but these are small platforms)
- Is there an incomplete platform with a very small DTS and still not compliant at all?
    - Yes!
        - More than one…
- Is there a platform with all targets failing (nothing warning-free)?

Quiz - SoC Platforms

- Is there a platform in the kernel fully compliant (warning-free)?
  - Yes!
    - More than one…
- Is there a platform fully compliant for all its architectures (e.g. ARM and ARM64)
  - Yes!
    - Amlogic
    - Actions and Airoha (but these are small platforms)
- Is there an incomplete platform with a very small DTS and still not compliant at all?
  - Yes!
    - More than one…
- Is there a platform with all targets failing (nothing warning-free)?
  - Yes!
    - More than one…

Thank you

Krzysztof Kozlowski
krzk@kernel.org, @krzk@social.kernel.org

Thank you